



US012398915B2

(12) **United States Patent**
Oh et al.

(10) **Patent No.:** **US 12,398,915 B2**

(45) **Date of Patent:** **Aug. 26, 2025**

(54) **PORTABLE AIR PURIFIER AND FAN
MODULE PROVIDED THEREIN**

(71) Applicant: **LG ELECTRONICS INC.**, Seoul
(KR)

(72) Inventors: **Si Young Oh**, Seoul (KR); **Seok-Ho
Choi**, Seoul (KR); **Sang Hyuk Son**,
Seoul (KR); **Juhyun Kim**, Seoul (KR)

(73) Assignee: **LG ELECTRONICS INC.**, Seoul
(KR)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 488 days.

(21) Appl. No.: **17/793,114**

(22) PCT Filed: **Apr. 23, 2020**

(86) PCT No.: **PCT/KR2020/005392**

§ 371 (c)(1),

(2) Date: **Jul. 15, 2022**

(87) PCT Pub. No.: **WO2021/149872**

PCT Pub. Date: **Jul. 29, 2021**

(65) **Prior Publication Data**

US 2023/0045895 A1 Feb. 16, 2023

(30) **Foreign Application Priority Data**

Jan. 22, 2020 (KR) 10-2020-0008743

(51) **Int. Cl.**

F24F 13/20 (2006.01)

F24F 3/16 (2021.01)

F24F 13/28 (2006.01)

(52) **U.S. Cl.**

CPC **F24F 13/20** (2013.01); **F24F 3/16**
(2013.01); **F24F 13/28** (2013.01); **F24F**
2221/12 (2013.01)

(58) **Field of Classification Search**

CPC B01D 46/00; B01D 46/0043; B01D
46/0056; B01D 46/69; B01D 46/2403;
B01D 46/42; B60H 1/00; B60H 3/06
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,507,422 B2 * 12/2019 Combs B01D 46/429
2017/0087500 A1 3/2017 Combs et al.
2019/0060817 A1 2/2019 Park et al.

(Continued)

FOREIGN PATENT DOCUMENTS

CN 203454279 2/2014
CN 205468438 8/2016

(Continued)

OTHER PUBLICATIONS

Korean Office Action issued in Application No. 10-2020-0008743
dated Sep. 19, 2024.

(Continued)

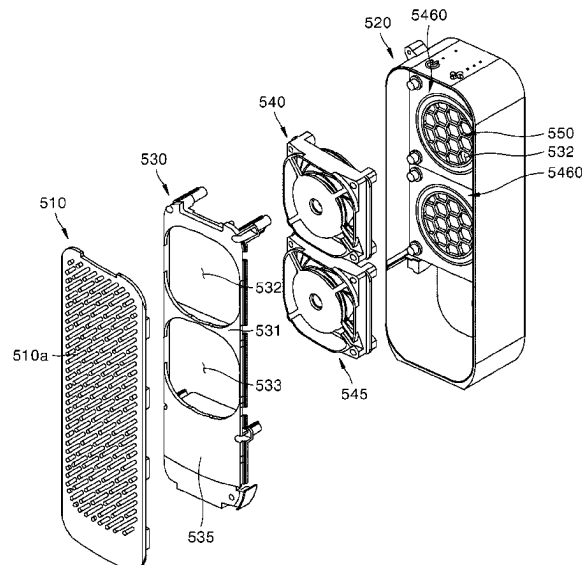
Primary Examiner — Sharon Pregler

(74) *Attorney, Agent, or Firm* — KED & ASSOCIATES

(57) **ABSTRACT**

A portable air purifier and a fan module therein are provided.
The portable air purifier includes a fan module that is
disposed between a discharge surface and a filter, and
suctions air having passed through the filter, in an axial
direction, and discharges the air in a direction between the
axial direction and a radial direction.

3 Claims, 17 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

2020/0282797 A1 9/2020 Kim et al.
2020/0284449 A1 9/2020 Kim et al.

FOREIGN PATENT DOCUMENTS

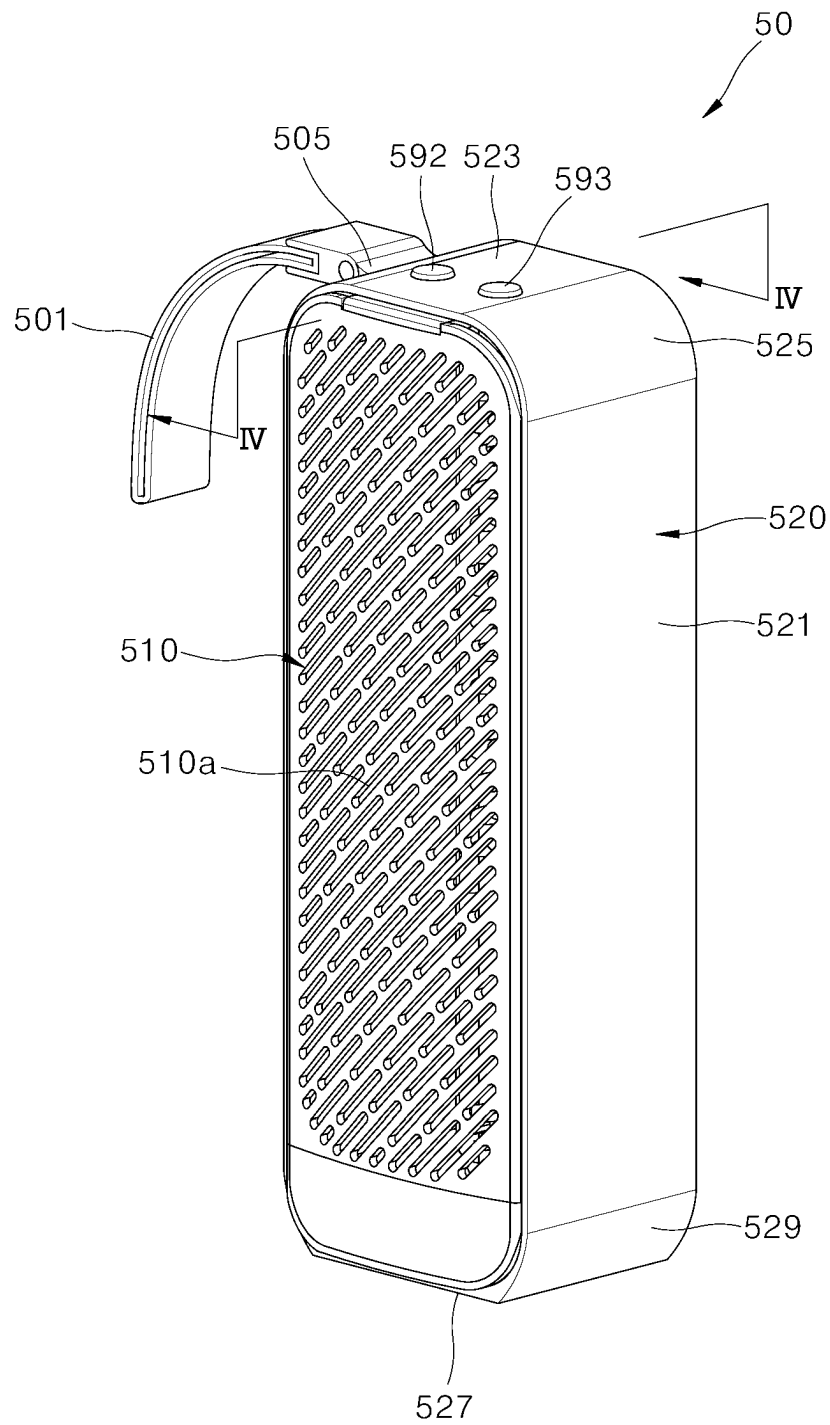
CN	206540257	10/2017	
JP	2015-045308	3/2015	
KR	20-0475454	12/2014	
KR	10-2019-0029478	3/2019	
KR	10-1926727	3/2019	
KR	10-1965793	4/2019	
KR	10-2019-0117066	10/2019	
WO	WO-2012081929 A2 *	6/2012	 F24F 8/10
WO	WO 2017/053979	3/2017	

OTHER PUBLICATIONS

German Office Action dated Dec. 1, 2022.
International Search Report dated Oct. 12, 2020 issued in Application No. PCT/KR2020/005392.

* cited by examiner

FIG. 1



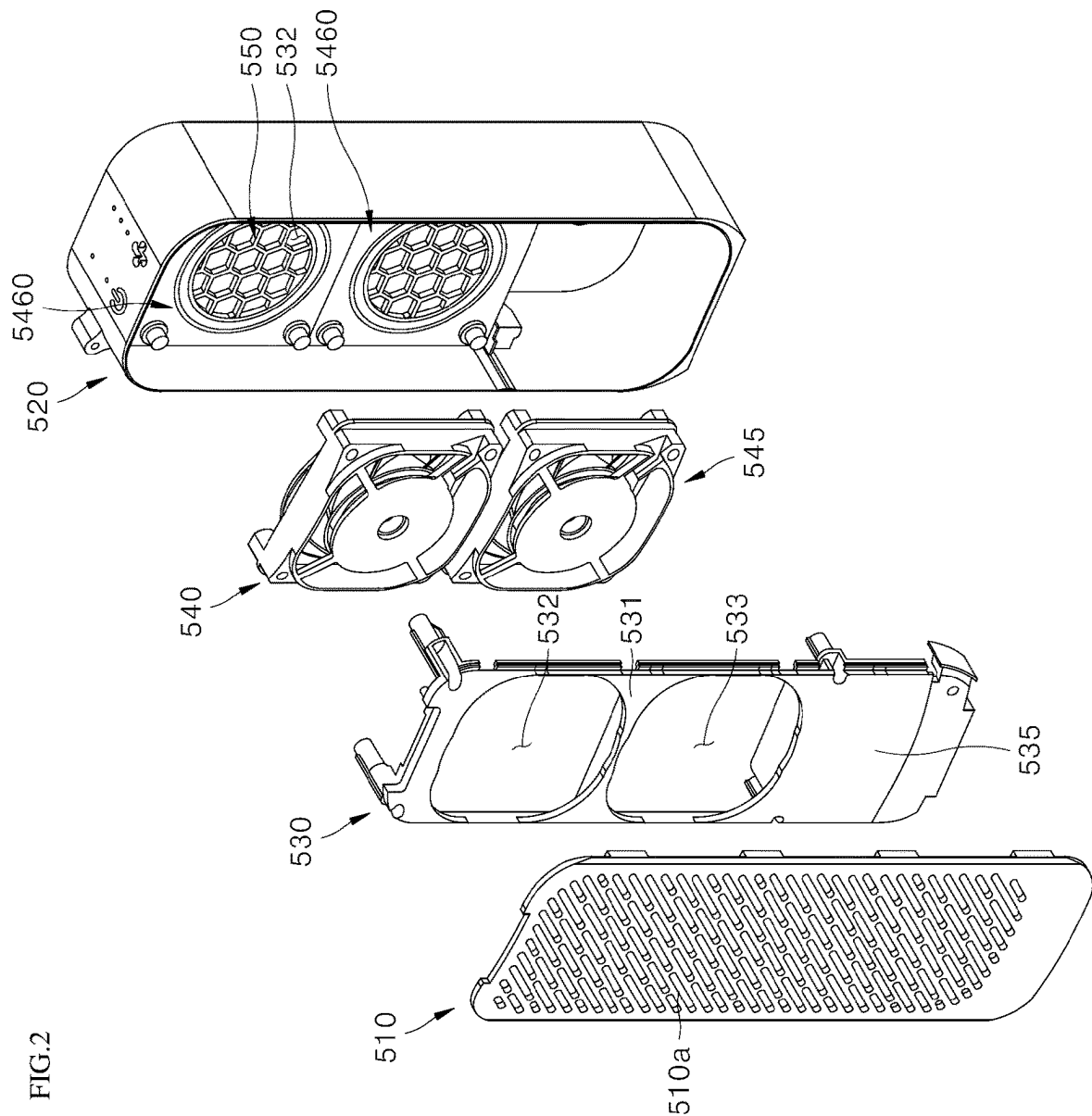


FIG. 3

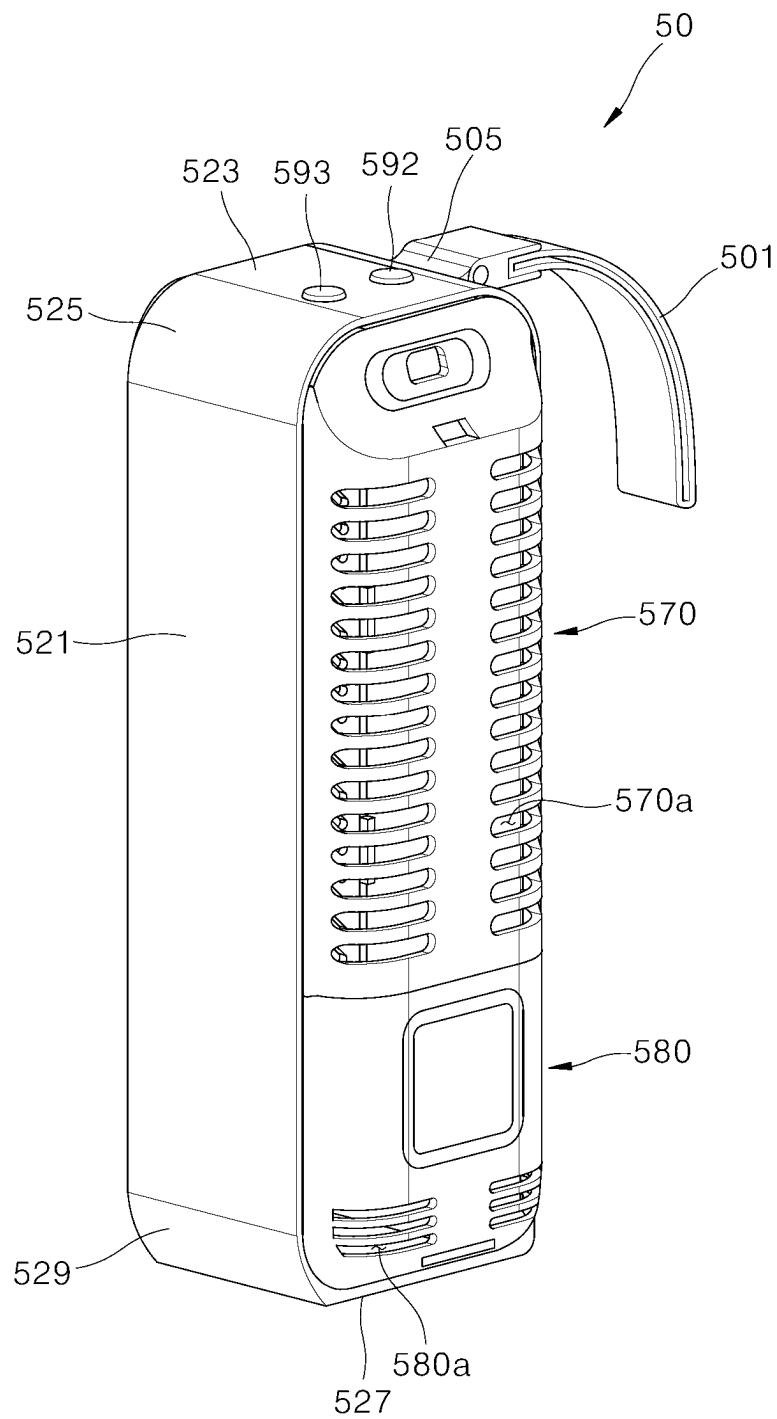


FIG. 4

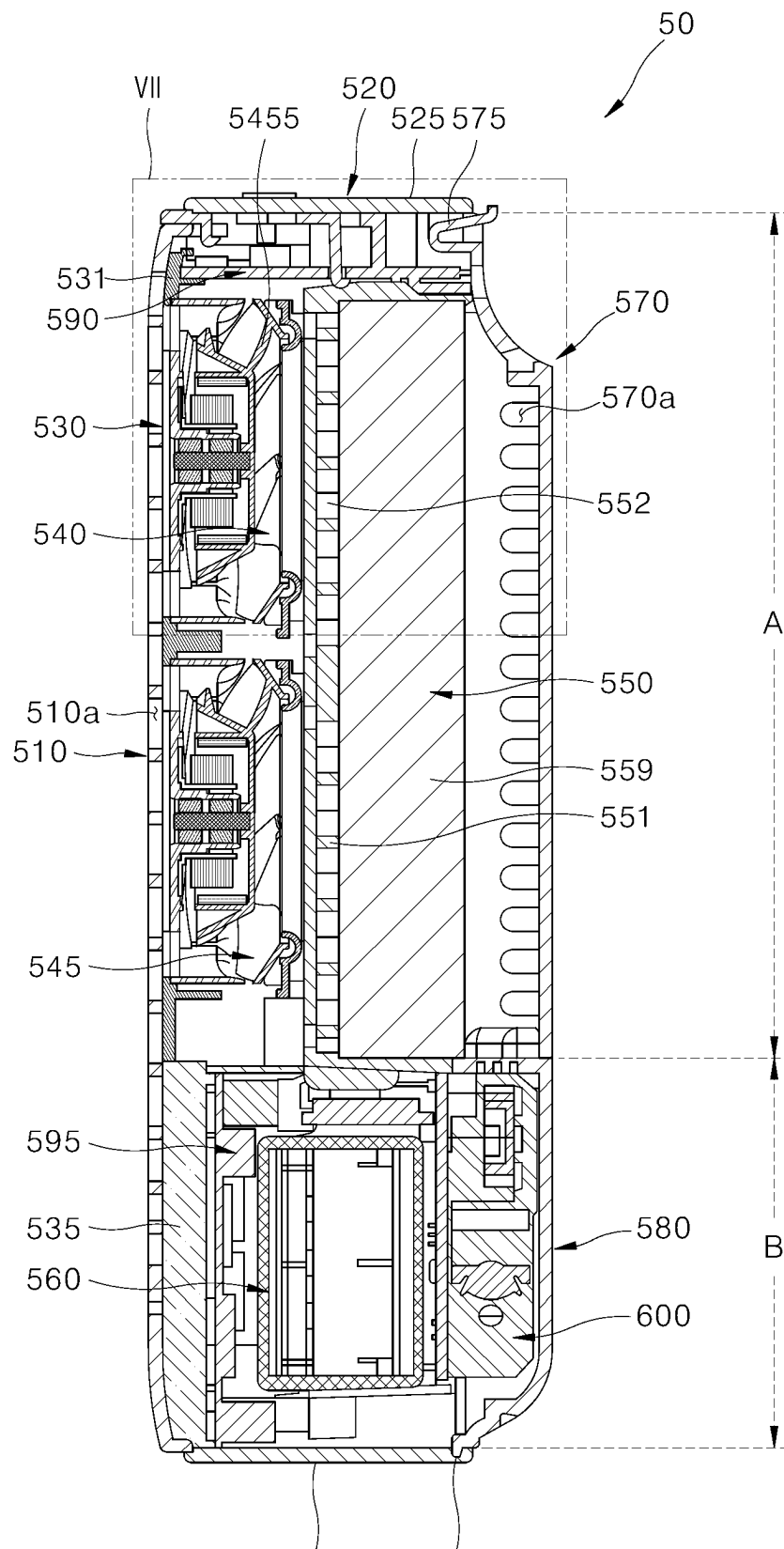


FIG. 5

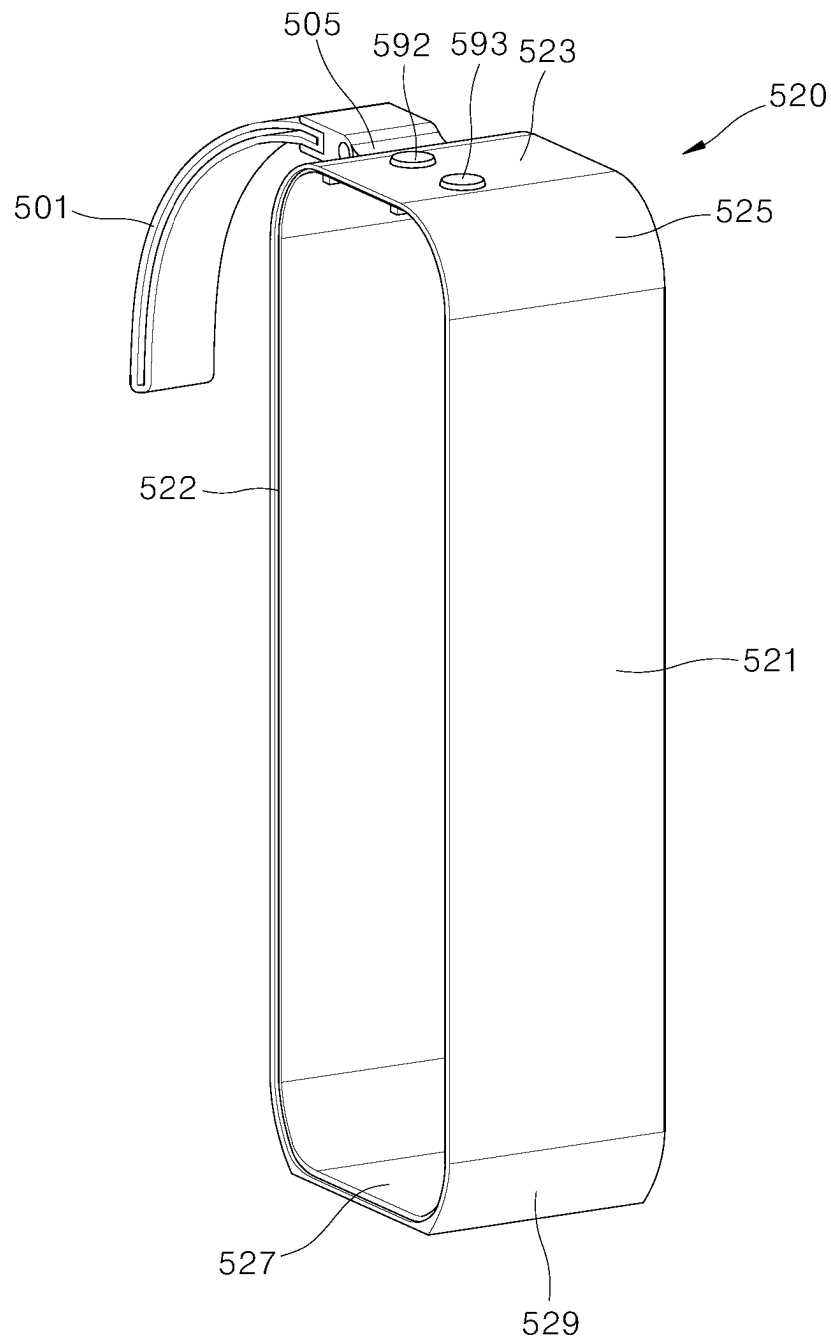


FIG. 6

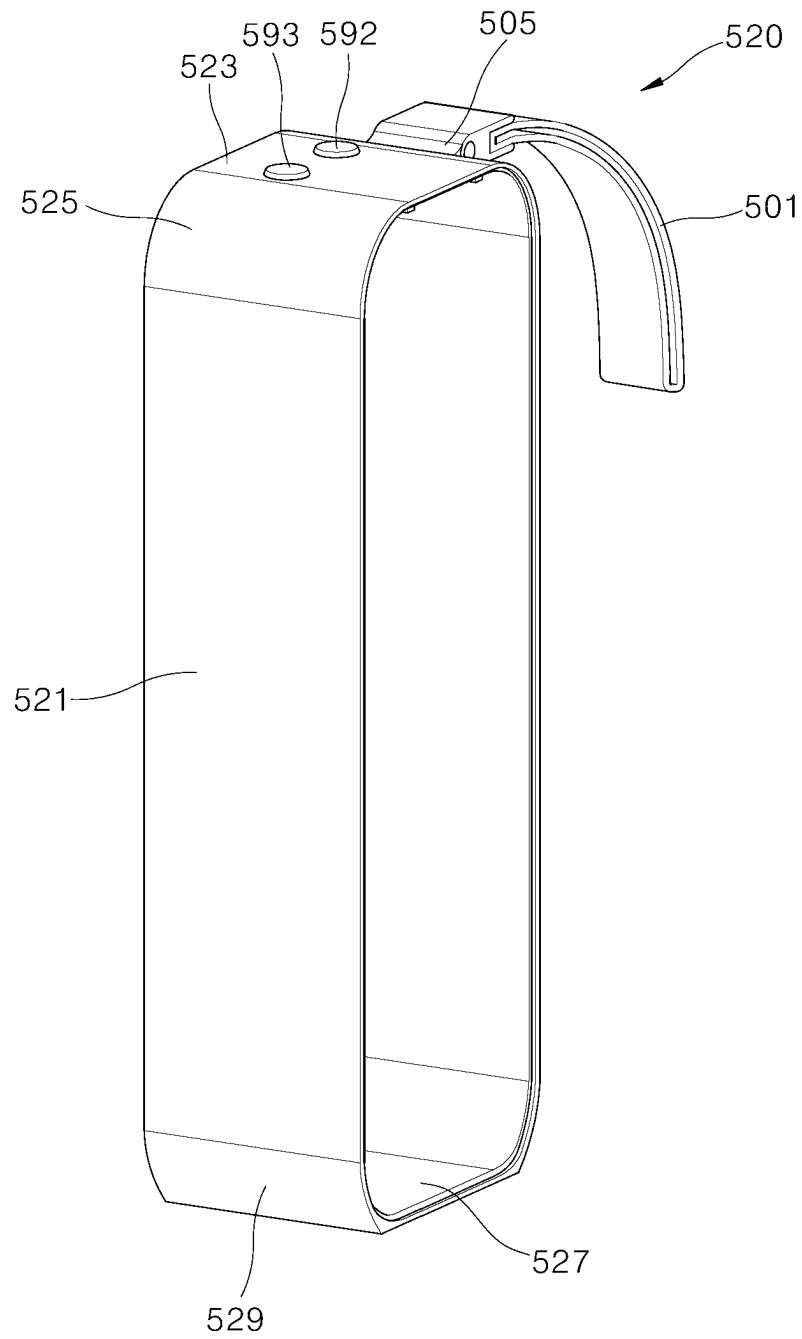


FIG. 7

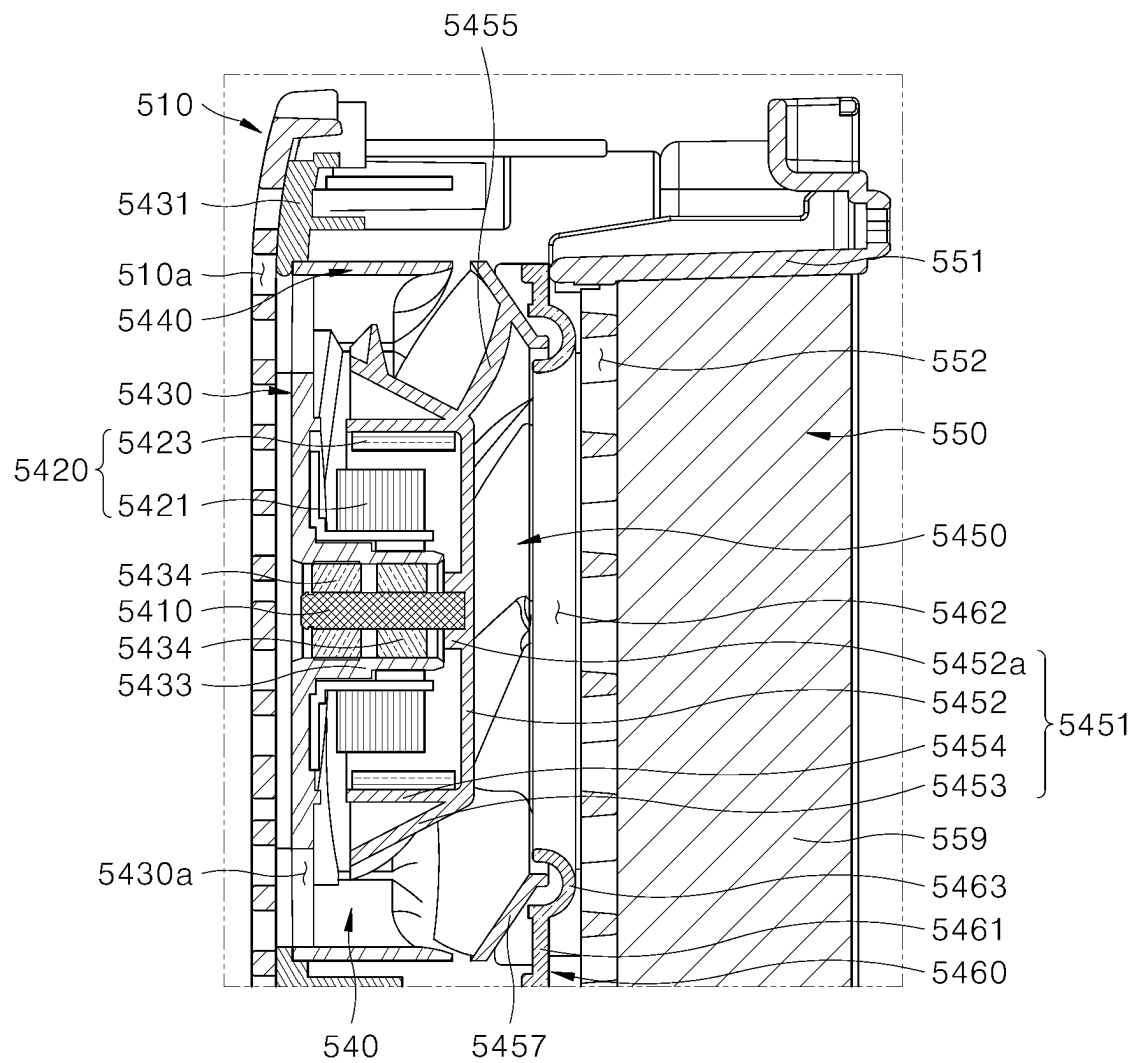


FIG. 8

540

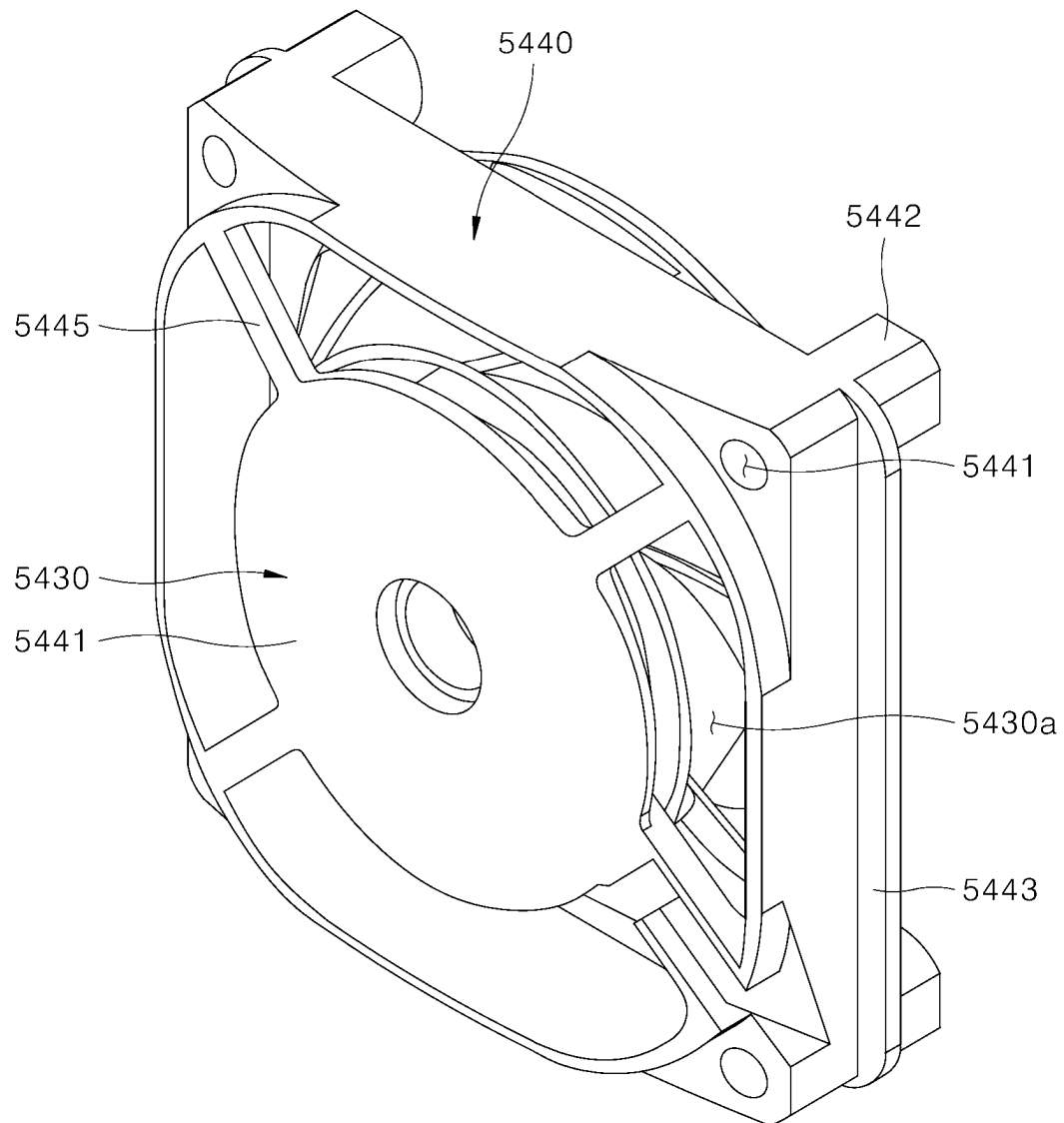


FIG. 9

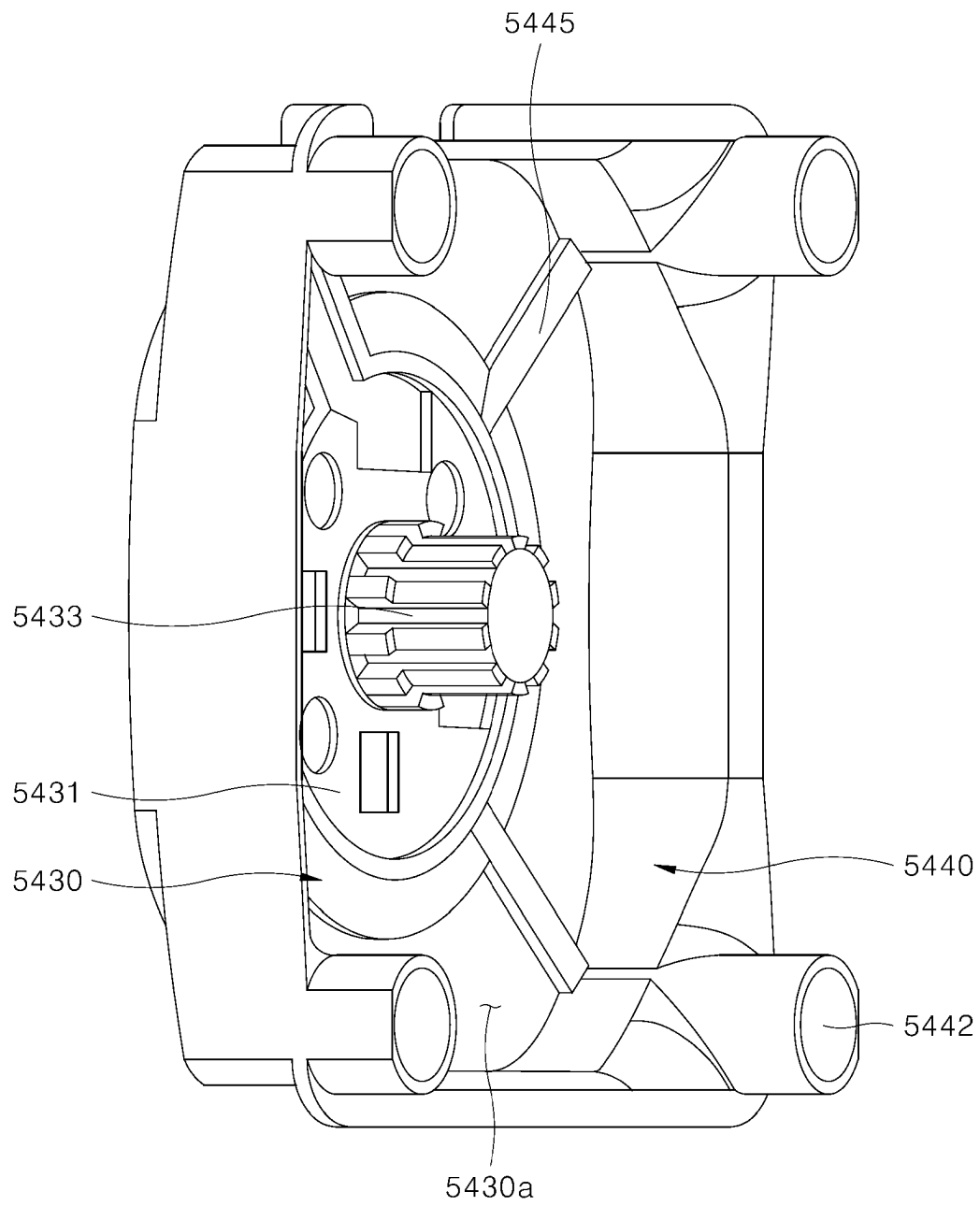


FIG. 10

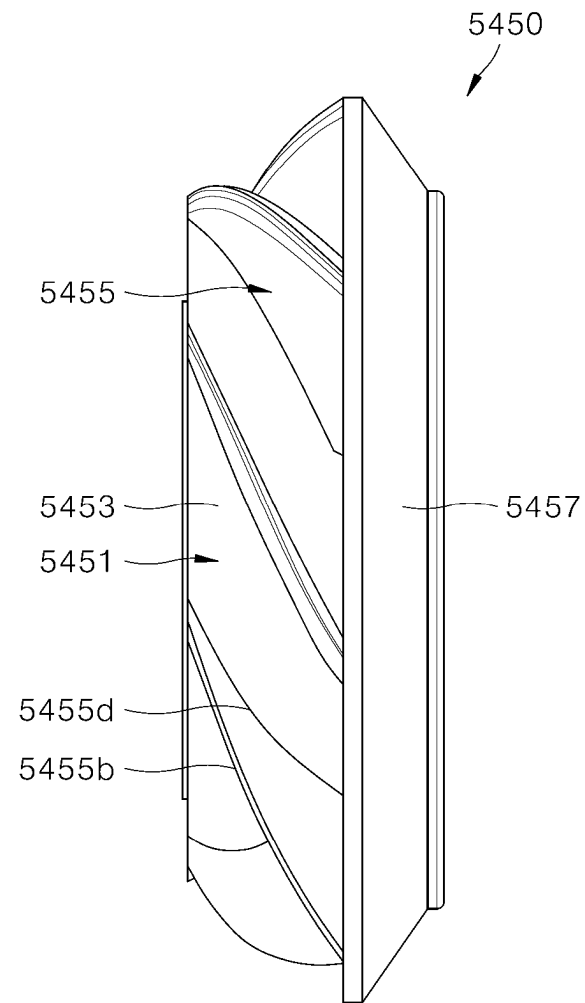


FIG. 11

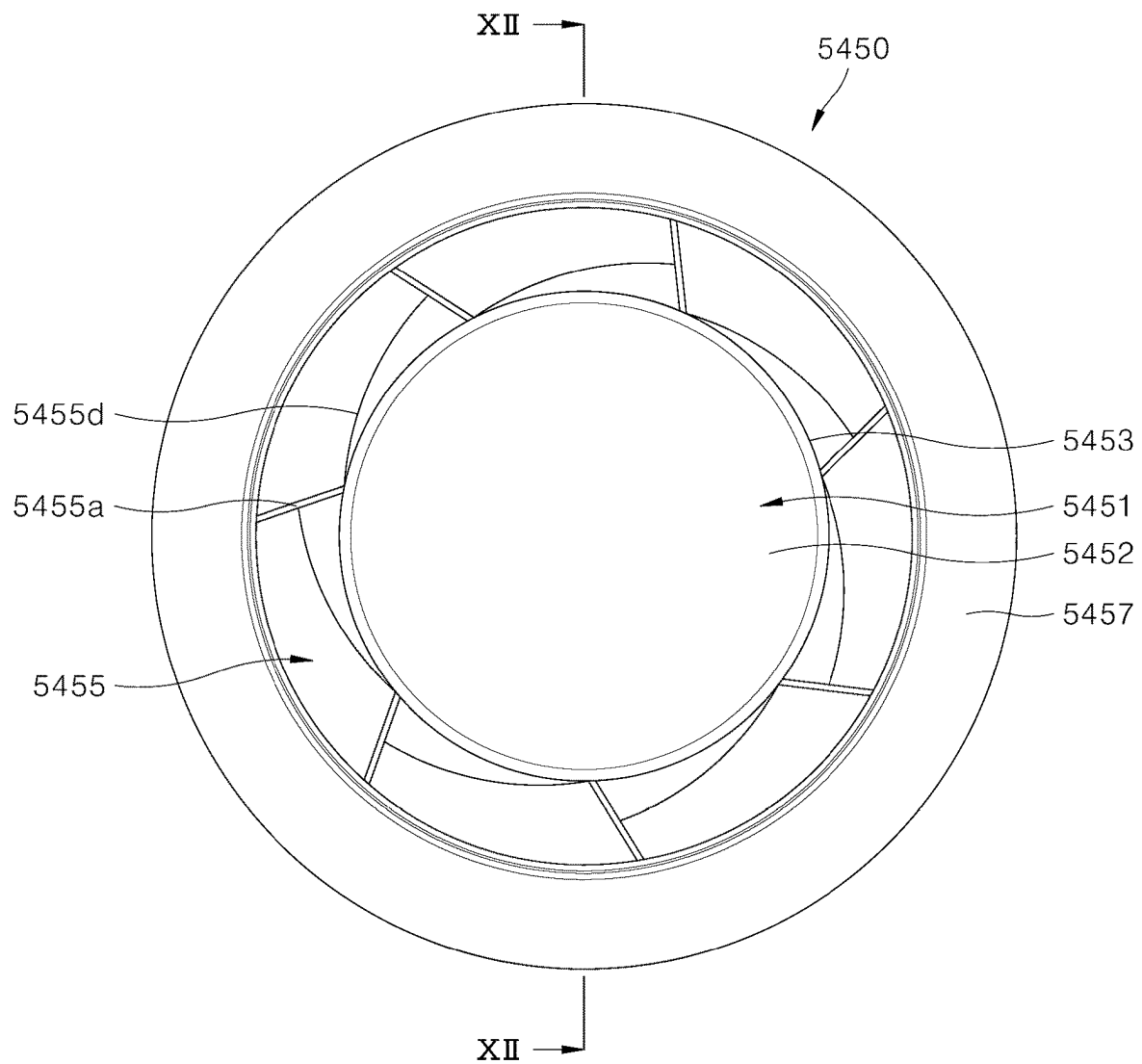


FIG.12

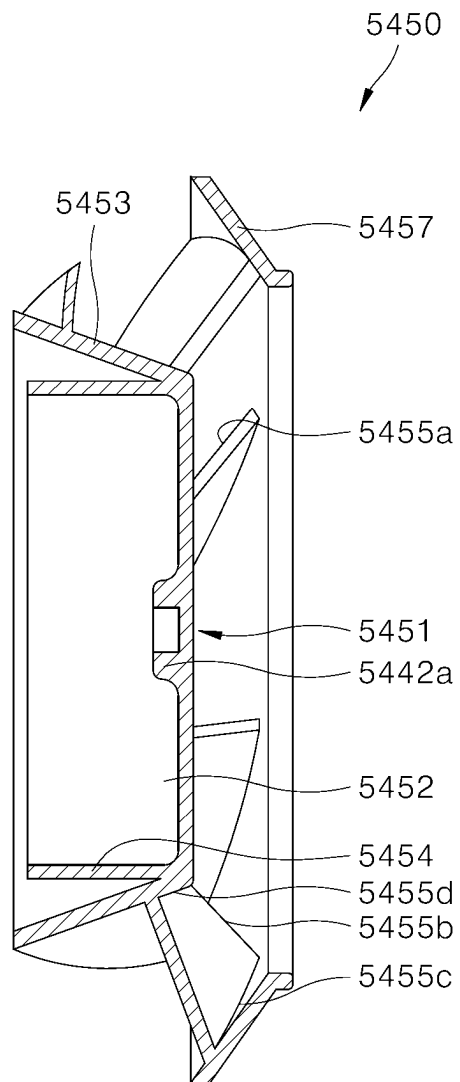


FIG. 13

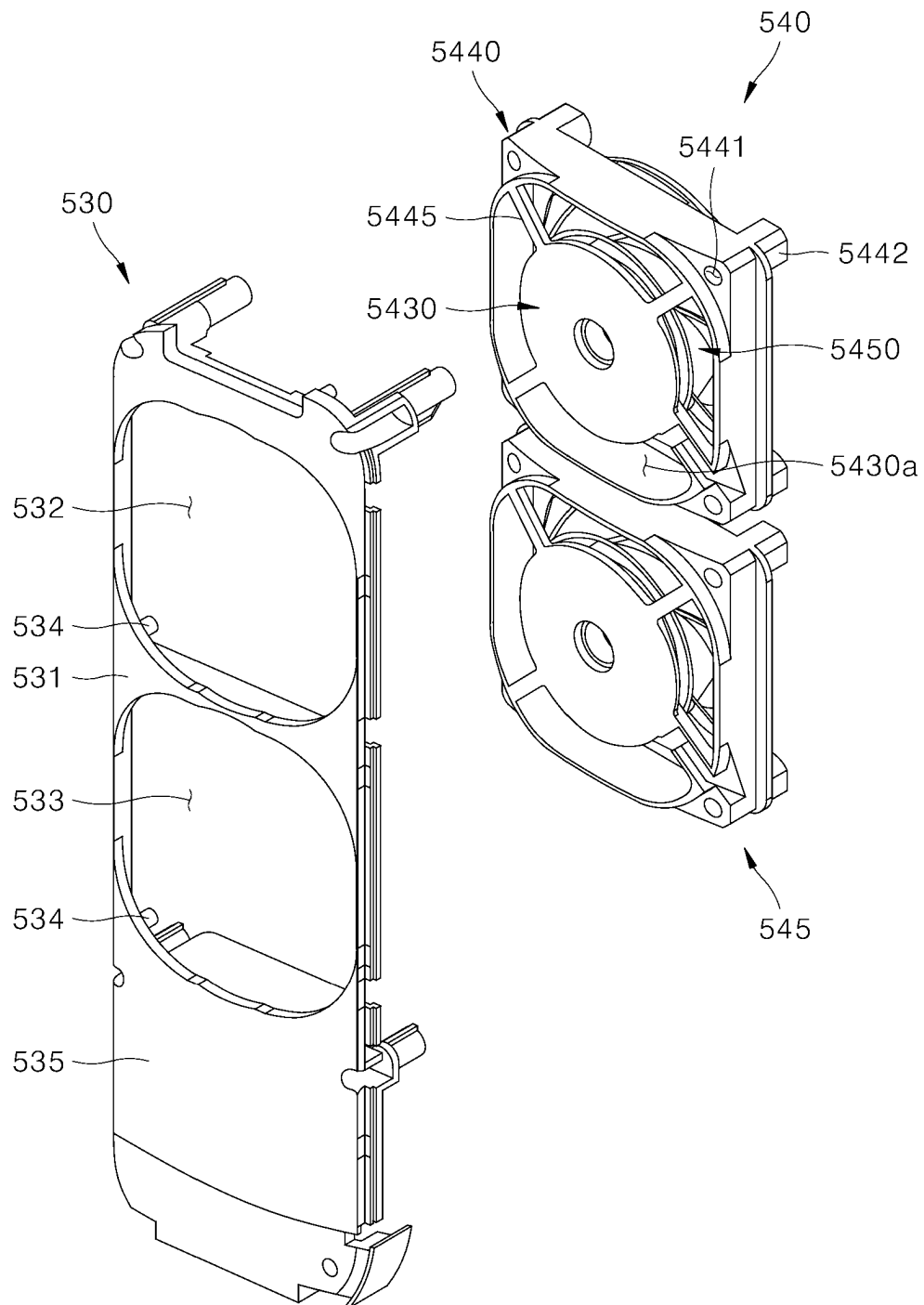


FIG. 14

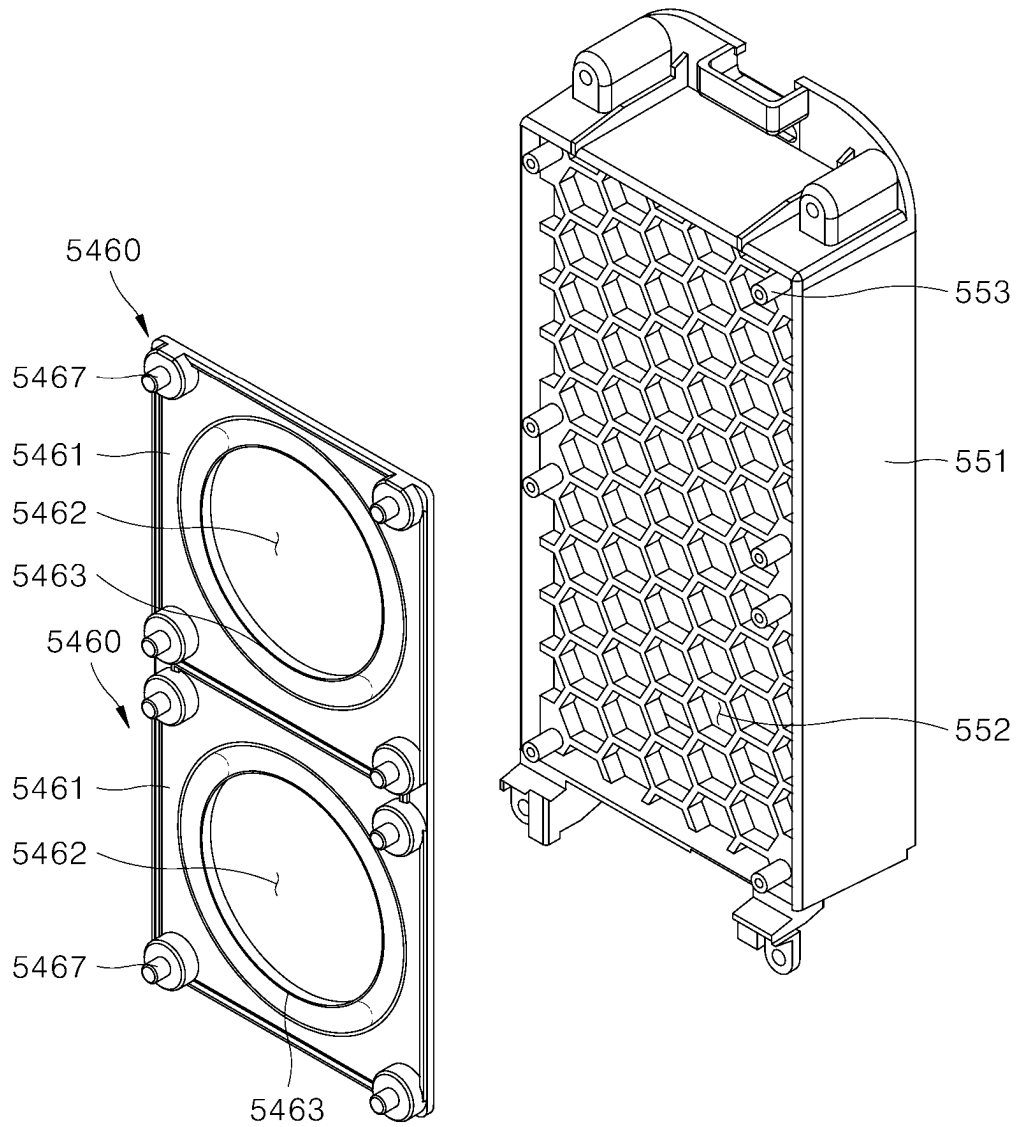


FIG. 15

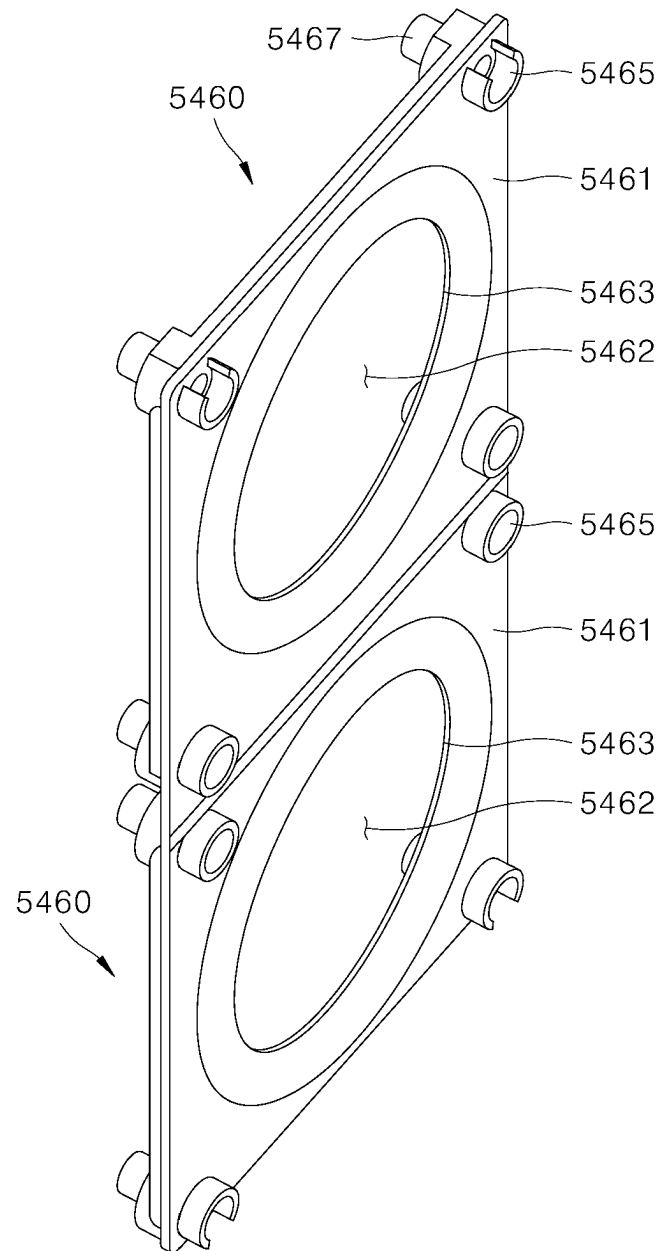


FIG. 16

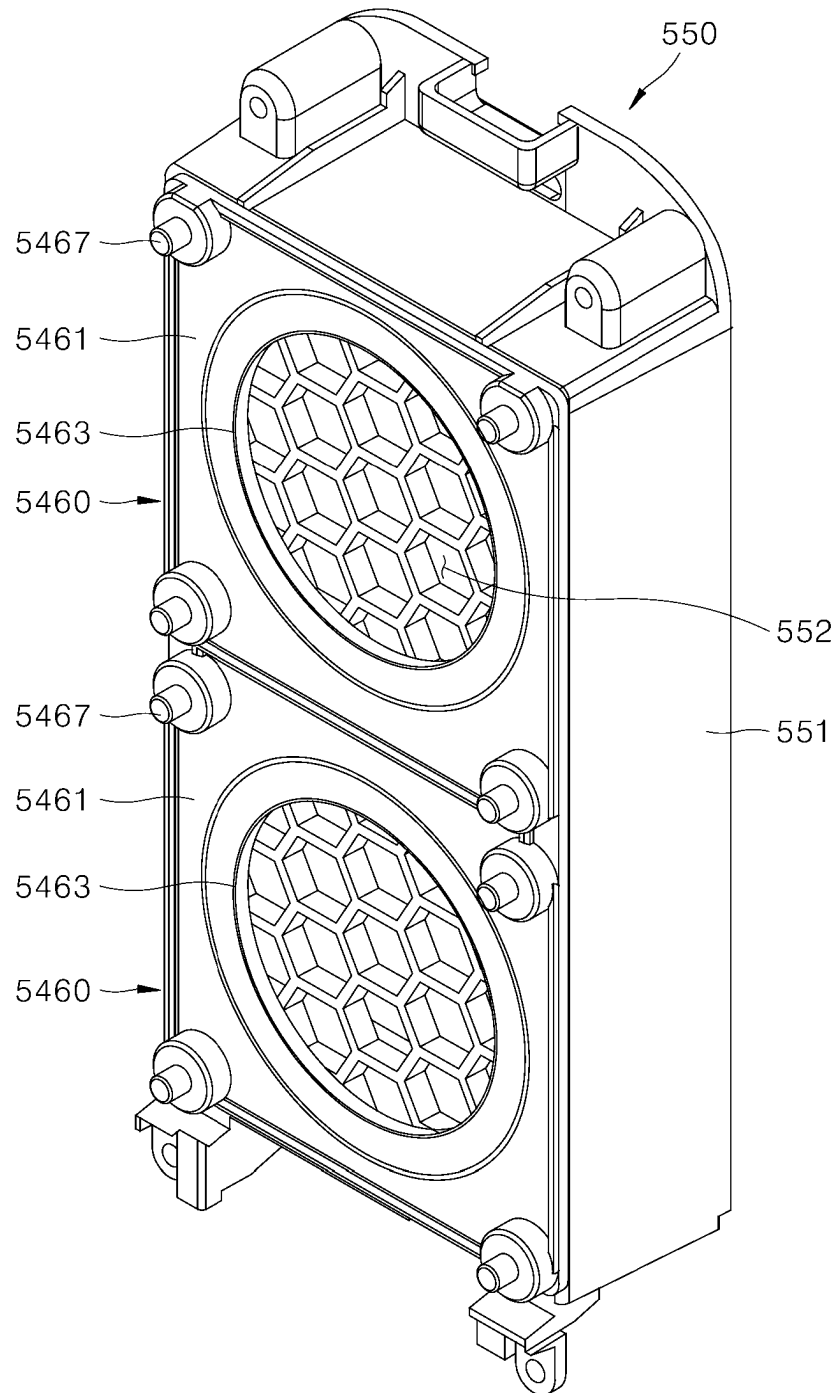
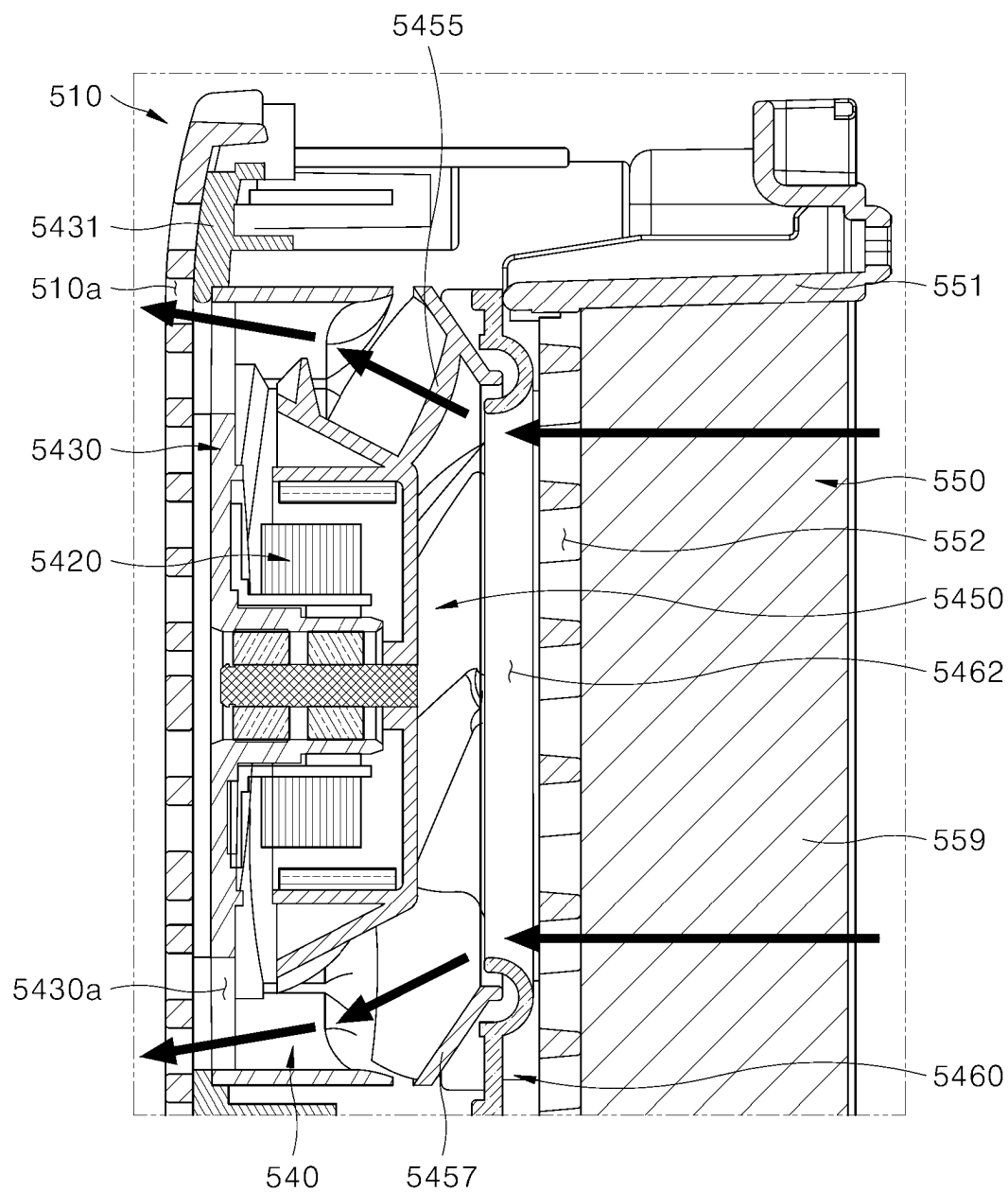


FIG. 17



PORTABLE AIR PURIFIER AND FAN MODULE PROVIDED THEREIN

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS

This application is a U.S. National Stage Application under 35 U.S.C. § 371 of PCT Application No. PCT/KR2020/005392, filed Apr. 23, 2020, which claims priority to Korean Patent Application No. 10-2020-0008743, filed Jan. 22, 2020, whose entire disclosures are hereby incorporated by reference.

TECHNICAL FIELD

Disclosed herein is a portable air purifier, and specifically, are a portable air purifier that can be carried for use, and a fan module provided in the portable air purifier.

BACKGROUND ART

Air purifiers are widely used in our daily lives. The devices can filter physical particles such as dust, fine dust, ultra fine dust and the like, chemical substances such as odorant particles, harmful gases and the like, and microorganisms such as germs, viruses and the like, in air, to purify the air.

People cannot live without air purifiers in an industrial society since more and more people are greatly affected by fine dust and suffer from allergies. Accordingly, there is a growing demand for the device.

Ordinarily, a large-sized air purifier is used in a house that is 100 square meters or greater. The air purifier can be provided with a filter for physical particles such as dust and the like, a filter for chemical substances such as gas and the like, and a filter for microorganisms such as germs, viruses and the like, in combination. That is, such a large-sized air purifier capable of accommodating various types of filters can be used in a large space.

However, air purifiers are rarely used in a narrow space such as a studio apartment, a space in a vehicle and the like, or in a very wide space such as a public library and the like or an outdoor space, considering space availability, portability and energy efficiency. Additionally, a user who moves from place to place usually uses an air purifier small enough to carry. Under the circumstances, there is a growing need for a portable air purifier that is easy to carry for use.

Portable air purifiers need to be small and lightweight enough for users to carry such that the users can easily carry and use the portable air purifiers anywhere. That is, the devices are useful for people who often go out and move from place to place instead of staying in a place such as a house.

The amount of purified air discharged from a portable air purifier is less than the amount of purified air discharged from an ordinary air purifier that is installed in a place. Additionally, the portable air purifier discharges purified forward in a narrower range than the ordinary air purifier. The portable air purifier has the above problems inherently, since the portable air purifier has a small size. As the size of the portable air purifier decreases, it is difficult to increase the amount of discharge of purified air and the range of forward discharge of purified air.

A decrease in the amount of discharge of purified air and the range of forward discharge of purified air of the portable air purifier makes it difficult for air purified by the portable air purifier to reach a user, in particular, a user's face. The

portable air purifier's failure in reaching the user's face means that the portable air purifier cannot provide air purification performance properly.

DESCRIPTION OF INVENTION

Technical Problems

One objective of the present disclosure is to provide a portable air purifier and a fan module provided in the portable air purifier that is compact and lightweight and has an improved structure enabling purified air to reach a user's face effectively.

Another objective of the present disclosure is to provide a portable air purifier and a fan module provided in the portable air purifier that ensures excellent air purification performance at fixed pressure.

Yet another objective of the present disclosure is to provide a portable air purifier and a fan module provided in the portable air purifier that has a compact size and ensures excellent air purification performance although a mixed flow fan having a greater length in an axial direction than an axial flow fan is applied to the portable air purifier.

Technical Solutions

According to one aspect, a portable air purifier according to one aspect includes a fan module that is disposed between a discharge surface and a filter, and suctions air having passed through the filter, in an axial direction, and discharges the air in a direction between the axial direction and a radial direction.

According to another aspect, a suction surface, a fan module and a discharge surface are disposed in a straight line, and the fan module disposed between the suction surface and the discharge surface suctions air in the axial direction and discharges the air in a direction between the axial direction and the radial direction.

Accordingly, the portable air purifier can discharge purified air farther away than an axial flow fan to a wider area, and ensure excellent air purification performance at fixed pressure.

According to another aspect, a fan blade connects to a skirt part that is formed at a slant, and a rotor of a motor is fixed to an inner projection part protruding between the skirt part and a stator.

The skirt part may have a truncated cone shape, the inner projection part may be formed into a cylinder having a hollow hole, and the rotor may be fixed to the inner circumferential surface of the inner projection part.

Thus, the rotor may be reliably fixed to the inside of a fan, and an increase in the weight of the fan is suppressed, improving the rigidity of the entire fan.

According one aspect of the present disclosure, a portable air purifier may include: a case having a suction surface on one side thereof in an axial direction and having a discharge surface on the other side thereof in the axial direction; a filter being disposed between the suction surface and the discharge surface; and a fan module being disposed between the discharge surface and the filter, and suctioning air, having passed through the filter, in the axial direction, and discharging the air in a direction between the axial direction and a radial direction.

According to the present disclosure, the portable air purifier may further include a fan base being disposed between the filter and the fan module, and the fan base

3

includes a fan inlet forming a passage that connects between the filter and the fan module, for example.

The fan base further includes a base plate being disposed between the filter and the fan module, and a bell mouth protruding from the base plate toward the fan module in the axial direction, and the fan inlet is formed at the inside of the bell mouth in a way that penetrates in a radial direction of the bell mouth, for example.

The fan module includes a motor including a stator and a rotor that is disposed outside the stator in a radial direction of the stator; a fan case supporting at least a portion of the motor; and a fan rotating together with the rotor, and the fan case is coupled to the fan base and the fan base is coupled to the filter, for example.

The fan case includes a first support part supporting the stator and the shaft; a second support part supporting the first support part while surrounding the first support part from the outside in the radial direction; and a projection boss being provided at the second support part, and being coupled to the fan base, for example.

The fan base includes a base plate being disposed between the filter and the fan module, and a fastening projection protruding from the base plate toward the fan module in the axial direction, for example.

The projection boss protrudes from the second support part toward the base plate in the axial direction, for example.

Additionally, at least a portion of the fan protrudes outward in a radial direction of the second support part, and as the projection boss and the fastening projection are coupled, the fan base and the fan module are coupled such that the second support part and the base plate are spaced a predetermined distance apart from each other, and the predetermined distance is a length or greater of the portion of the fan, protruding outward in the radial direction of the second support part, in the axial direction, for example.

The portable air purifier further includes a fan cover being disposed between the discharge surface and the fan module, and the fan cover is provided with an air discharge part forming a passage that connects between the discharge surface and the fan, and the fan module is disposed in a position facing the air discharge part, for example.

The case has a length extending in a first direction, a width extending in a second direction orthogonal to the first direction, and a height extending in a third direction orthogonal to the first direction and the second direction, and the inner space of the case, formed between the suction surface and the discharge surface, is divided into a first area at one side in the third direction and a second area at the other side in the third direction, and the air discharge part is disposed in the first area, for example.

According to the present disclosure, the portable air purifier further includes a battery disposed in the second area, and the fan cover further includes a lower cover part disposed in the second area, and the battery is disposed between the lower cover part and the suction surface, for example.

The direction in which the fan suctions air is the same as the directions in which the suction surface and the discharge surface are disposed, for example.

The suction surface is disposed in parallel with the discharge surface, for example.

The filter includes a filter surface that forms a flat surface in a direction orthogonal to the direction in which the fan suctions and discharges air, and the suction surface, the filter and the discharge surface are disposed in a straight line, for example.

4

A plurality of fan modules is disposed between the discharge surface and the filter, and the surface area taken up by the plurality of fan modules corresponds to the surface area of the discharge surface, for example.

The fan module includes a first fan module being disposed in parallel with the discharge surface, between the discharge surface and the filter; and a second fan module being disposed on the same surface as the first fan module, for example.

The case has a length extending in a first direction, a width extending in a second direction orthogonal to the first direction, and a height extending in a third direction orthogonal to the first direction and the second direction, and the positions of the first fan module and the second fan module in the third direction do not overlap each other, for example.

The direction in which the first fan module suctions and discharges air, and the direction in which the second fan module suctions and discharges air are the same, and the first fan module and the second fan module are disposed in the third direction, for example.

The filter includes a filter surface disposed in parallel with at least any one of the suction surface and the discharge surface, and the positions of the first fan module and the second fan module in the first and second directions correspond to the area occupied by the filter surface, for example.

The surface areas taken up by the first fan module and the second fan module correspond to the surface area of the filter surface, for example.

The fan module includes a mixed flow fan, for example.

According to another aspect of the present disclosure, a fan module may include: a shaft extending in an axial direction; a motor including a stator and a rotor that is disposed outside the stator in a radial direction of the stator; a fan case supporting the stator and the shaft; and a fan including a hub that rotates together with the rotor and the shaft, and a fan blade that protrudes from the hub in a centrifugal direction, the hub may include: a hub plate part being disposed in parallel with the support plate, with the boss therebetween, and having a circular plate shape; a skirt part protruding from an edge of the hub plate part toward the support plate, being disposed outside the stator in the radial direction of the stator, and forming a slant surface that inclines in the centrifugal direction as the skirt part becomes farther from the hub plate part in the axial direction; and an inner projection part protruding from the hub plate part toward the support plate, and being disposed between the skirt part and the stator, and the fan blade may protrude from the skirt part in the centrifugal direction, and the rotor may be disposed between the inner projection part and the stator and fixed to the inner projection part.

The fan case includes a first support part supporting the stator and the shaft; and a second support part supporting the first support part while surrounding the first support part from the outside in the radial direction, for example.

The first support part includes a support plate having a circular plate shape, and a boss protruding from the support plate toward the fan in the axial direction, the stator is coupled to the boss while surrounding the boss from the outside in the radial direction, and the support plate supports the boss coupled to the stator, for example.

The boss has a hollow hole therein, and the shaft is inserted into the boss, passes through the boss and connects to the hub, for example.

A bearing is inserted into the boss, and the shaft is rotatably supported by the bearing, for example.

5

The second support part is spaced a predetermined distance apart from the support plate in the centrifugal direction, a fan outlet is formed between the support plate and the second support part, and the fan case further includes a connection part that connects between the support plate and the second support part and helps the second support part to support the support plate, for example.

A plurality of connection parts is spaced at a predetermined interval along a circumferential direction of the support plate, and the fan outlet is respectively formed among the connection parts, for example.

The fan further includes a shroud being disposed outside the hub in a radial direction of the hub in a way that the shroud is spaced a predetermined distance apart from the hub, and the fan blade connects between the hub and the shroud, for example.

The blade includes a leading edge **5455a** being disposed at a front end in a rotation direction and being formed in a straight manner; a trailing edge being disposed at a rear end in the rotation direction and being formed in a straight manner; a shroud chord **5455c** connecting one end of the leading edge **5455a** and one end of the trailing edge, and extending from an inner circumferential surface of the shroud; and a hub chord **5455d** connecting the other end of the leading edge **5455a** and the other end of the trailing edge, and extending from an outer circumferential surface of the hub, for example.

The shroud forms a slant surface that inclines in a centripetal direction as the shroud becomes farther from the fan case in the axial direction, the skirt part forms a slant surface that inclines in the centrifugal direction as the skirt part becomes farther from the hub plate in the axial direction, one end of the leading edge **5455a** and one end of the trailing edge connect to the inner circumferential surface of the shroud, and the other end of the leading edge **5455a** and the other end of the trailing edge connect to an outer circumferential surface of the skirt part, for example.

Further, one end of the leading edge **5455a** is disposed closer to the center of the hub plate in a radial direction of the hub plate than one end of the trailing edge, and the other end of the leading edge **5455a** is disposed closer to the center of the hub plate in the radial direction of the hub plate than the other end of the trailing edge, for example.

Advantageous Effects

A portable air purifier according to the present disclosure is provided with a fan having a similar weight and size to an axial flow fan and capable of discharging purified air farther away than the axial flow fan to a wider area. The portable air purifier is compact and lightweight, and allows purified air to reach a user's face effectively, ensuring improvement in air purification performance.

The portable air purifier according to the present disclosure is provided with a fan that is advantageous over an axial flow fan in the discharge of high-pressure high-speed air, ensuring excellent suction and discharge performance despite the application of a high performance filter, and excellent air purification performance at fixed pressure.

The portable air purifier according to the present disclosure is provided with a fan that helps to discharge high-pressure high-speed air far away without a structure such as a guide vane, such that the size of the fan increases or the size of the portable air purifier decreases as much as the space occupied by the guide vane decreases.

According to the present disclosure, the fan protrudes further in a centrifugal direction than the inner circumfer-

6

ential surface of the fan case, surrounding the fan, such that the size of the fan increases within in a limited space in the portable air purifier.

Thus, according to the present disclosure, air purification performance improves further, with no increase in the size of the portable air purifier, caused by the fan.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a front perspective view showing the front surface side of a portable air purifier of one embodiment.

FIG. 2 is an exploded perspective view showing a state in which the portable air purifier in FIG. 1 is exploded.

FIG. 3 is rear perspective view showing the rear surface side of the portable air purifier in FIG. 1.

FIG. 4 is a cross-sectional view along line "IV-IV" in FIG. 1.

FIG. 5 is a front perspective view separately showing a case in FIG. 1.

FIG. 6 is a rear perspective view separately showing the case in FIG. 1.

FIG. 7 is a cross-sectional view specifically showing the structure of portion "VII" in FIG. 4.

FIG. 8 is a perspective view separately showing a fan module of one embodiment.

FIG. 9 is a perspective view separately showing a fan case in FIG. 8.

FIG. 10 is a front view separately showing the fan in FIG. 8.

FIG. 11 is a side view showing the fan in FIG. 10.

FIG. 12 is a cross-sectional view along line "XII-XII" in FIG. 11.

FIG. 13 is an exploded perspective view separately showing a fan cover and a fan module.

FIG. 14 is an exploded perspective view separately showing a fan module, a fan base and a filter.

FIG. 15 is a rear perspective view showing the rear surface of a fan base.

FIG. 16 is a perspective view showing a coupling state between the fan base and the filter.

FIG. 17 is a view showing an aspect of the air flow of the portable air purifier of one embodiment.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

The above-described aspects, features and advantages are specifically described hereunder with reference to the accompanying drawings such that one having ordinary skill in the art to which the present disclosure pertains can easily embody the technical spirit of the disclosure. In the disclosure, detailed description of known technologies in relation to the subject matter of the disclosure is omitted if it is deemed to make the gist of the disclosure unnecessarily vague. Below, preferred embodiments according to the disclosure are specifically described with reference to the accompanying drawings. In the drawings, identical reference numerals can denote identical or similar components.

The terms "first", "second" and the like are used herein only to distinguish one component from another component. Thus, the components should not be limited by the terms. Certainly, a first component can be a second component unless stated to the contrary.

When one component is described as being "in the upper portion (or the lower portion)" or "on (or under)" another component, one component can be directly on (or under)

another component, and an additional component can be interposed between the two components.

When any one component is described as being “connected”, “coupled”, or “connected” to another component, any one component can be directly connected or coupled to another component, but an additional component can be “interposed” between the two components or the two components can be “connected”, “coupled”, or “connected” by an additional component.

Throughout the disclosure, each component can be provided as a single one or a plurality of ones, unless stated to the contrary.

In the disclosure, singular forms include plural forms as well, unless explicitly indicated otherwise. It is to be understood that the terms such as “comprise” or “include” and the like, when used in this disclosure, are not interpreted as necessarily including stated components or steps, but can be interpreted as excluding some of the stated components or steps or as further including additional components or steps.

Throughout the disclosure, the terms “A and/or B” as used herein can denote A, B or A and B, and the terms “C to D” can denote C or greater and D or less, unless stated to the contrary.

[Exterior of Portable Air Purifier]

FIG. 1 is a front perspective view showing the front surface side of a portable air purifier of one embodiment, and FIG. 2 is an exploded perspective view showing a state in which the portable air purifier in FIG. 1 is exploded. FIG. 3 is rear perspective view showing the rear surface side of the portable air purifier in FIG. 1, and FIG. 4 is a cross-sectional view along line “IV-IV” in FIG. 1.

In the embodiment, a portable air purifier 50 is described as an example.

Referring to FIGS. 1 to 4, the portable air purifier 50 in the embodiment may be approximately formed into a cuboid. The portable air purifier 50 includes a case 520, a front surface panel 510 and a rear surface panel 570.

The case 520 forms the skeleton of the exterior of the portable air purifier 50. Various types of components are accommodated in the case 520.

Both sides of the case 520 in a first direction may be open. That is, the front and the rear of the case 520 may be open. A suction surface may be disposed on one side of the case 520 in the first direction, and a discharge surface may be disposed on the other side of the case 520 in the first direction.

For example, the suction surface may be disposed at the rear of the case 520, and the rear surface panel 570 may be disposed at the rear of the case 520, where the suction surface is disposed. Additionally, the discharge surface may be disposed at the front of the case 520, and the front surface panel 510 may be disposed at the front of the case 520, where the discharge surface is disposed.

The suction surface may indicate a virtual surface corresponding to the boundary between the inside of the case 520 and the outside of the case 520, at the rear of the case 520. The discharge surface indicates a virtual surface corresponding to the boundary between the inside of the case 520 and the outside of the case 520, at the front of the case 520.

In the embodiment, the suction surface and the discharge surface may actually be openings formed at the rear and the front of the case 520, since both the rear and the front of the case 520 are open. In the embodiment, the suction surface and the discharge surface are disposed in parallel, for example.

The front surface panel 510 is coupled to the front of the case 520. The front surface panel 510 forms the exterior of the front surface of the portable air purifier 50.

The rear surface panel 570 is coupled to the rear of the case 520. The rear surface panel 570 forms the exterior of the rear surface of the portable air purifier 50.

The portable air purifier 50 may be entirely formed into a standing cuboid that is elongated in the up-down direction. Accordingly, a user may use the portable air purifier 50 in the state in which the portable air purifier 50 stands or lies. Additionally, the portable air purifier 50 may stay in the same position reliably without rolling in a moving vehicle, even if the portable air purifier 50 is used in the state in which the portable air purifier 50 lies.

Directions are defined as follows. A direction from the case 520 to the front surface panel 510 is referred to as a front, and a direction from the case 520 to the rear surface panel 570 is referred to as a rear. At this time, a “first direction” denotes the front-rear direction. Additionally, the first direction may have the same meaning as the axial direction. The axial direction may be defined as a parallel direction with the lengthwise direction of a shaft disposed at a fan module 540 that is described below. A “second direction” is a direction perpendicular to the first direction and denotes the left-right direction. A “third direction” is a direction perpendicular to the first direction and the second direction and denotes the up-down direction.

[Entire Structure of Portable Air Purifier]

The portable air purifier 50 in the embodiment includes a front surface panel 510, a case 520, a fan cover 530, a fan module 540, 545, a filter module 550, a battery 560, a rear surface panel 570 and a rear surface cover 580.

The front surface panel 510 is disposed at the front most side of the portable air purifier 50 and forms the exterior of the front surface of the portable air purifier 50. Air purified by the portable air purifier 50 is discharged outward through the front surface panel 510. To this end, a plurality of outlets 510a is provided on the front surface panel 510.

The case 520 forms the skeleton of the exterior of the portable air purifier 50. The exteriors of the upper surface, lateral surfaces and lower surface of the portable air purifier 50 are formed by the case 520. Accommodation space is formed in the case 520. Various types of components such as a fan cover 530, a fan module 540, 545, a battery 550, a filter module 560 and the like, constituting the portable air purifier 50, are accommodated in the accommodation space. The case 520 has enough strength to protect the accommodated components from an external impact, for example.

The fan cover 530 is accommodated in the accommodation space of the case 520 and disposed at the front of the fan module 540, 545. That is, the fan cover 530 is disposed between the front surface panel 510 and the fan module 540, 545, inside the case 520.

The fan cover 530 fixes the fan module 540, 545 to the inside of the case 520. Additionally, the fan cover 530 also induces air being blown by the fan module 540, 545 to flow straight toward the front, without causing the spread of the air to surrounding areas. Further, the fan cover 530 may be involved in fixing the filter module 550 and the battery 560.

The fan module 540, 545 may be accommodated in the accommodation space of the case 520, and disposed between the discharge surface and the filter module 550. Specifically, the fan module 540, 545 may be disposed between the fan cover 530 and the filter module 550. That is, the fan module 540, 545 is disposed at the rear of the fan cover 530 and the front of the filter module 550. The fan

module **540**, **545** suctions air from the rear of the portable air purifier **50** and discharges air to the front of the portable air purifier **50**.

In the embodiment, the fan module **540**, **545** includes a mixed flow fan respectively, for example. The fan module **540**, **545** may suction air having passed through the filter module **550** in the axial direction and discharge the air in a direction between the axial direction and the radial direction.

The filter module **550** is accommodated in the accommodating space of the case **520** and disposed between the fan module **540**, **545** and the rear surface panel **570**. That is, the filter module **550** is disposed at the rear of the fan module **540**, **545** and the front of the rear surface panel **570**.

The filter module **550** purifies air suctioned through the rear of the portable air purifier **50**. The air that is purified while passing through the filter module **550** passes through the fan module **540**, **545**, the fan cover **530** and the front surface panel **510** and then is discharged from the front of the portable air purifier **50**.

The filter module **550** may include a filter case **551** and a filter **559**.

The filter case **551** forms the skeleton of the exterior of the filter module **550**. In the embodiment, the filter case **551** is formed into a cuboid, the rear surface of which is open, for example. Insertion space for accommodating the filter **559** is formed in the filter case **551**. The rear of the filter case **551** is open. Accordingly, a passage for inserting the filter **559** into insertion space in a case main body part **552** is formed.

The filter **559** is mounted in the insertion space of the filter case **551**. The filter case **551** may be provided with a mounting groove or a mounting projection that allows the filter **559** to be firmly mounted in the filter case **551**, on the inner surface thereof.

Additionally, the filter case **551** is provided with a penetration hole **552** that forms a passage between the insertion space and the fan module **540**, **545**. The penetration hole **552** is formed in a way that penetrates in the front-rear direction on the front surface of the filter case **551**. The penetration hole **552** forms a passage for allowing air having passed through the filter **559** to flow toward the fan module **540**, **545**.

A plurality of penetration holes **552** is formed on the front surface of the filter case **551**, and each of the penetration holes **552** is formed in a way that penetrates in the front-rear direction on the front surface of the filter case **551**.

In the embodiment, a plurality of penetration holes **552** is provided on the front surface of the filter case **551**, and each penetration hole **552** is formed into a hexagon, for example. The plurality of penetration holes **552** is arranged in the form of a honeycomb. Accordingly, a honeycomb structure may be formed on the front surface of the filter case **551**.

The honeycomb structure, formed on the front surface of the filter case **551** as described above, ensures improvement in the rigidity of the filter case **551**, makes the filter case **551** lightweight as well as ensuring a passage for allowing air to flow.

The battery **560** is accommodated in the accommodation space of the case **520**, and disposed under the fan module **540**, **545** and the filter module **550**. The battery **560** can supply a power source for driving the portable air purifier **50**. To this end, the battery **560** may electrically connect to the fan module **540**, **545**, the filter module **550** and at least one of a sub PCB **590** and a main PCB **595** that are described hereafter.

Together with the rear surface cover **580**, the rear surface panel **570** is disposed at the rearmost side of the case **520** and forms the exterior of the rear surface of the portable air

purifier **50**. The rear surface panel **570** is disposed behind the filter module **560**. External air is suctioned into the portable air purifier **50** through the rear surface panel **570**. To this end, the rear surface panel **570** has a plurality of first inlets **570a**, thereon.

Additionally, together with the rear surface panel **570**, the rear surface cover **580** is disposed at the rearmost side of the case **520** and forms the exterior of the rear surface of the portable air purifier **50**. The rear surface cover **580** is disposed behind the battery **560**. In the embodiment, the area behind the filter module **550** is covered by the rear surface panel **570**. The area behind the battery **560** is covered by the rear surface cover **580**.

The rear surface cover **580** may have a second inlet **580a**. The second inlet **580a** is formed in a way that penetrates on the rear surface cover **580**. The second inlet **580a** forms a passage connecting among the rear of the case **520**, the battery **560** and a sensor module **600**. Through the second inlet **580a**, external air may flow to the battery **560** and the sensor module **600** in a second area B.

[Structure of Case]

FIG. **5** is a front perspective view separately showing a case in FIG. **1**, and FIG. **6** is a rear perspective view separately showing the case in FIG. **1**.

Referring to FIGS. **3**, **5** and **6**, the portable air purifier **50** includes a case **520** that forms the skeleton of the exterior of the portable air purifier **50**. The case **520** has accommodation space therein, and one side and the other side of the accommodation space in the first direction are open.

Directions are defined as follows. A direction from the case **520** to the front surface panel **510** is referred to as a front, and a direction from the case **520** to the rear surface panel **570** is referred to as a rear. At this time, a “first direction” denotes the front-rear direction. A “second direction” is a direction perpendicular to the first direction and denotes the left-right direction. A “third direction” is a direction perpendicular to the first direction and the second direction and denotes the up-down direction.

In the embodiment, the case **520** is formed into a cuboid having the front surface and the rear surface that are open, and therein, has accommodation space having the front and the rear that are open. The case **520** may be made of a metallic material. In the embodiment, the case **520** is made of a material such as aluminum that is lightweight and has high strength, for example.

The rear surface of the case **520** is open to suction external air. The front surface of the case **520** is open to discharge air purified in the accommodation space of the case **520**. Through the open front surface and the open rear surface of the case **520**, various types of components constituting the portable air purifier **50** may be installed in the accommodation space of the case **520**.

A filter module **550** may be mounted in the case **520** or detached from the case **520** through the open rear surface of the case **520**. The rear surface panel **570** is coupled to the open rear surface of the case **520**. The rear surface panel **570** coupled to the case **520** covers the open rear surface of the case **520**.

The case **520** may include a first surface part **521**, a second surface part **523**, a first connection surface part **525**, a third surface part **527**, and a second connection surface part **529**, while being formed into a cuboid having the front surface and the rear surface that are open.

The first surface part **521** forms a surface in a direction that is perpendicular to the second direction perpendicular to the first direction. That is, the first surface part **521** forms a lateral surface of the case **520**. The first surface part **521** is

formed into a vertical flat surface that forms a wall covering a lateral portion of the accommodation space in the case 520.

The case 520 has a pair of first surface parts 521, and the first surface parts 521 face each other and are spaced a predetermined distance apart from each other. At this time, the pair of first surface parts 521 is disposed in the second direction, i.e., a direction in which the first surface parts 521 are disposed side by side in the left-right direction.

The second surface part 523 forms a surface in a direction that is perpendicular to the third direction and is disposed at one side of the pair of first surface parts 521 in the third direction. That is, the second surface part 523 is disposed in the upper portion of the first surface part 521, and forms a flat surface in a direction parallel with the direction in which the first surface parts 521 are spaced, i.e., a horizontal flat surface. The second surface part 523 forms the upper surface of the case 520.

The first connection surface part 525 is disposed between the first surface part 521 and the second surface part 523. The first connection surface part 525 is disposed respectively between the end portion of one side of the second surface part 523 and the first surface part 521 below the end portion of one side of the second surface part 523, and between the end portion of the other side of the second surface part 523 and the first surface part 521 below the end portion of the other side of the second surface part 523.

Each of the first connection surface parts 525 connects between the first surface part 521 and the second surface part 523 in a rounded manner. The first connection surface part 525 makes an upper edge of the case 520, at which the first surface part 521 and the second surface part 523 connect, round, improving the safety of a product and the aesthetic qualities of the exterior of a product.

The third surface part 527 is disposed below the second surface part 523 and the first surface part 521, and forms a flat surface parallel with the second surface part 523. The third surface part 527 forms the lower surface of the case 520. Additionally, the third surface part 527 is a portion that supports the portable air purifier 50 such that the portable air purifier 50 keeps standing.

The second connection surface part 529 is disposed between the first surface part 521 and the third surface part 527. The second connection surface part 529 is disposed respectively between the end portion of one side of the third surface part 527 and the first surface part 521 over the end portion of one side of the third surface part 527, and between the end portion of the other side of the third surface part 527 and the first surface part 521 over the end portion of the other side of the third surface part 527.

Each of the first connection surface parts 525 connects between the first surface part 521 and the second surface part 523 in a rounded manner. The first connection surface part 525 makes an upper edge of the case 520, at which the first surface part 521 and the third surface part 527 connect, round, improving the safety of a product and the aesthetic qualities of the exterior of a product.

The case 520 may have a power button 592 in the upper portion thereof, i.e., in the second surface part 523 of the case 520. The power button 592 is provided as a manipulation button for turning on/off the portable air purifier 50.

Additionally, the case 520 may have an air volume control button 593, in the upper portion thereof. The air volume control button 593 is provided as a manipulation button for increasing or decreasing the air volume of the portable air purifier 50. The air volume control button 593 may be

disposed near the power button 592 such that the user identifies and manipulates the air volume control button 593 readily.

Further, the case 520 may have a strap mounting part 505 for coupling the strap 501 to the case 520. The strap 501 is provided to allow the user to hold the case 520. The user can move up the portable air purifier 50 by holding the strap 501 without directly holding the portable air purifier 50. That is, the strap 501 helps to improve the portability of the portable air purifier 50.

The accommodation space in the case 520 may be divided into a first area A and a second area B. When the accommodation space is divided in the up-down direction, the upper area is the first area A, and an area under the first area A is the second area B. The first area A and the second area B are divided conceptually rather than physically.

In the embodiment, among the components accommodated in the accommodation space, the fan modules 540, 545 and the filter module 550 are disposed in the first area A, and the battery 560 is disposed in the second area B.

[Disposition Structure of Components of Portable Air Purifier]

Referring to FIGS. 1 to 4, the accommodation space in the case 520 forming the skeleton of the portable air purifier 50 is divided into a first area A in the upper portion of the case 520 and a second area B in the lower portion of the case 520.

Components in relation to the suction, purification and discharge of air are disposed in the first area A. That is, the filter module 550 and the fan module 540, 545 are disposed in the first area A, and accordingly, air flows in the first area A. A plurality of first inlets 570a is provided as a passage for suctioning air, on the rear surface panel 570. An air discharge part 532, 533 and an outlet 510a are provided as a passage for discharging air purified in the first area A, on the fan cover 530 and the front surface panel 510. Additionally, a flow path connecting among the first inlet 570a, the air discharge part 532, 533 and the outlet 510a is formed in the first area A.

That is, the first inlet 570a, the filter module 550, the fan module 540, 545, the air discharge part 532, 533 and the outlet 510a are provided in the first area A, and a flow path for allowing air suctioned into the portable air purifier 50 to pass through the air purifier 50 is formed in the first area A.

Components that do not directly relate to an air flow for air purification are disposed in the second area B. That is, the main PCR 595, the battery 560 and the sensor module 600 are disposed in the second area B. The rear cover 580 covers the open rear of the space where the above components are disposed.

In the embodiment, the case 520 is formed in to a cuboid having a length in the up-down direction greater than a length in the lateral direction. Additionally, the up-to-down length of the first area A in the upper portion of the case 520 is greater than that of the second area B in the lower portion of the case 520. That is, when the portable air purifier 50 stands vertically, the first area A in the upper portion of the case 520 occupies more space than the second area B in the lower portion of the case 520.

A lower cover part 535 of the fan cover 530 is disposed at the frontmost side of the second area B. The battery 560 is disposed behind the lower cover part 535 while being disposed in the second area B. The fan module 540, 545 and the filter module 550 are disposed over the battery 560, and the rear surface cover 580 is disposed behind the battery 560. Additionally, the sensor module 600 may be disposed between the battery 560 and the rear surface cover 580.

That is, the upper boundary of the battery 560 is defined by the fan module 540, 545 and the filter module 550, the lateral and lower boundaries of the battery 560 are defined by the first surface part 521 and the third surface part 527 of the case 520, and the rear boundary of the battery 560 is accommodated in space defined by the rear surface cover 580.

In the embodiment, the battery 560 is a heavier object than the fan module 540, 545 and the filter module 550. It is preferable that the battery 560 weights more than a total weigh of the fan module 540, 545 and the filter module 550.

Ordinarily, since the weight per unit volume of the battery 560 is much greater than that of the fan module 540, 545 and that of the filter module 550, the battery 560 may be readily provided as a heavier object than the fan module 540, 545 and the filter module 550 although the weight or size of the battery 560 does not increase intentionally.

That is, even if a battery 560 having capacity required for the ordinary use of an portable air purifier 50 is applied to the portable air purifier 50, naturally, the battery 560 weights more than the fan module 540, 545 and the filter module 550.

When the battery 560 as a heavy object is disposed in the lower portion of the portable air purifier 50, the following effects can be produced.

First, when the battery 560 that is a heavy object is disposed in the lower portion of the portable air purifier 50, the center of gravity of the portable air purifier 50 is biased toward the lower side of the portable air purifier 50 from the up-to-down center of the portable air purifier 50. That is, the center of gravity of the portable air purifier 50 is biased toward the lower side of the portable air purifier 50, in which the battery 560 is disposed.

The bias of the center of gravity of the portable air purifier 50 toward the lower portion of the portable air purifier 50, where the battery 560 is disposed, reduces the risk of the overturning of the portable air purifier 50, when the portable air purifier 50 stands vertically.

That is, when the portable air purifier 50 stands vertically, the portable air purifier 50 rarely falls since the center of gravity of the portable air purifier 50 is at the lower side of the portable air purifier 50 because of the battery 560 in the lower portion of the portable air purifier 50.

When it comes to the portable air purifier 50 in the embodiment, the third surface part 527 of the case 520 forms a flat surface having a greater surface area than the second surface part 523 of the case 520. That is, the third surface part 527 forms a long flat surface that has a greater length in the lateral direction than the second surface part 523. That is, under the assumption that the length of the third surface part 527 in the lateral direction is d1 and that the length of the upper surface part 527 in the lateral direction is d2, the relationship $d1 > d2$ is satisfied. Additionally, the second connection surface part 529 connecting between the third surface part 527 and the first surface part 521 forms a curved surface that has a less R value than the first connection surface part 525.

The third surface part 527 of the case 520 is a portion contacting the bottom surface of the portable air purifier 50 when the portable air purifier 50 stands vertically. That is, the third surface part 527 is a portion that supports the portable air purifier 50 such that the portable air purifier 50 keeps sanding.

Since the third surface part 527 of the case 520 forms a surface area that has a greater surface area than the second

surface part 523, the third surface part 527 of the case 520 can support the portable air purifier 50 that stands vertically, more reliably.

In combination of the structure in which the battery 560 is disposed in the lower portion of the portable air purifier 50 such that the center of gravity of the portable air purifier 50 is at the lower side of the portable air purifier 50, and the structure in which the third surface part 527 of the case 520, which supports the portable air purifier 50 standing vertically, forms a flat surface that has a greater surface area than the second surface area 523, effectively suppressing the overtraining of the portable air purifier 50 and reliably keeping the portable air purifier 50 standing vertically.

Second, when the battery 560 as a heavy object is disposed in the lower portion of the portable air purifier 50, the other components such as the filter module 550 and the fan module 540, 545 constituting the portable air purifier 50 need to be disposed further upward than the battery 560. That is, components in relation to the suction, purification and discharge of air need to be disposed further upward than the battery 560.

To ensure the charge capacity of the battery 560, required for the smooth operation of the portable air purifier 50, the battery 560 needs to have a predetermined size or greater. It means that the portable air purifier 50 needs to have installation space of a predetermined size or greater, therein, to install the battery 560. Additionally, it is irrational to form a flow path for allowing an air to flow in the space where the battery 560 is installed. Accordingly, components in relation to the suction, purification and discharge of air needs to be disposed to avoid the battery 560, i.e., in a position higher than the position of the battery 560.

In the disposition structure, a flow path for the suction, purification and discharge of air is formed in the first area A higher than the position of the battery 560, in the portable air purifier 50. Thus, the suction of air into the portable air purifier 50, and the discharge of air purified in the portable air purifier 50 are performed in a position higher than the position of the battery 560.

As purified air is discharged from the upper portion of the portable air purifier 50, the air purified in the portable air purifier 50 can reach the user's face more easily.

When the portable air purifier 50 is placed and used on the bottom surface lower than the user's face, the portable air purifier 50 that stands vertically allows a greater amount of air purified in the portable air purifier 50 to reach the user's face than the portable air purifier 50 that lies horizontally.

To this end, when purified air is discharged from the upper portion of the portable air purifier 50 in the state in which the portable air purifier 50 stands vertically, a greater amount of the air purified in the portable air purifier 50 reaches the user's face.

In the embodiment, since the battery 560 is disposed in the lower portion of the portable air purifier 50, a flow path for the suction, purification and discharge of air is formed in a position higher than the position of the battery 560, in the portable air purifier 50. Accordingly, purified air is discharged from the upper portion of the portable air purifier 50, and a greater amount of the air purified in the portable air purifier 50 reaches the user's face.

That is, in the structure where the battery 560 is disposed in the lower portion of the portable air purifier 50, the structural stability of the portable air purifier 50 can improve to reduce the risk of overturning of the portable air purifier 50 that stands vertically, and an efficient flow path can also be formed to enable a greater amount of air purified in the portable air purifier 50 to reach the user's face.

15

Third, the structure, in which the battery **560** as a heavy object is disposed in the lower portion of the portable air purifier **50** such that components in relation to the suction, purification and discharge of air are disposed in a position higher than the position of the battery **560**, may help to expand the range in which the portable air purifier **50** is installed.

In an example, when the portable air purifier **50** is used in the state of being held in a cup holder **h** in a vehicle, the area where air is suctioned and the area where purified air is discharged are disposed higher than the cup holder, such that the portable air purifier **50** is reliably held in vehicle while maintaining a high level of air purification performance. To this end, the up-to-down length of the second area **B** where the battery **560** is disposed is set to the depth of the cup holder or greater, for example.

In another example, if the lower area of the portable air purifier **50** is fixed by a tong type holder and the like, the portable air purifier **50** can be fixed stably while the portable air purifier **50**'s areas where air is suctioned and discharged are not blocked.

That is, components such as a battery **560** that does not directly relate to an air flow for air purification are disposed in the lower portion of the portable air purifier **50**, and the portable air purifier **50** is held and fixed through its lower portion, ensuring a high level of air purification performance and a reliable fixation of the portable air purifier **50**.

[Disposition Structure of Fan Module and Filter Module]

The portable air purifier **50** in the embodiment may include a fan module **540**, **545** that suction air from the rear of the portable air purifier **50** and discharges air through the front of the portable air purifier **50**.

The fan module **540**, **545** may be disposed between the fan cover **530** and the filter module **550**. That is, the fan cover **530** may be disposed at the front of the fan module **540**, **545**, and the filter module **550** may be disposed at the rear of the fan module **540**, **545**.

While the fan module **540**, **545** is accommodated in the accommodation space of the case **520**, the fan module **540**, **545** may be disposed in the first area **A**. Accordingly, the fan module **540**, **545** may be disposed in a position that faces the rear surface of the upper cover part **531**, out of the upper cover part **531** of the fan cover **530** and the lower cover part **535** of the fan cover **530**.

In the embodiment, the portable air purifier **50** is provided with two fan modules **540**, **545**, for example. At this time, the portable air purifier **50** is provided with a first fan module **540** and a second fan module **545**.

The first fan module **540** may be disposed between the discharge surface and the filter module **550**, specifically, between the fan cover **530** and the filter module **550**. At this time, the first fan module **540** may be disposed in parallel with the suction surface and the discharge surface. The second fan module **545** and the first fan module **540** may be disposed on the same surface. That is, the second fan module **545** may be disposed in parallel with the suction surface and the discharge surface, between the fan cover **530** and the filter module **550**.

The positions of the first fan module **540** and the second fan module **545** are disposed not to overlap each other in the third direction. To this end, the first fan module **540** and the second fan module **545** may be disposed in the third direction, i.e., in the up-down direction. That is, the first fan module **540** and the second fan module **545** may be disposed on the same perpendicular line.

The first fan module **540** and the second fan module **545** may be disposed to face the air discharge part **532**, **533**

16

formed on the fan cover **530**. Accordingly, air, suctioned into the first fan module **540** and the second fan module **545** through the rear surface panel **570**, may pass through an upper air discharge part **532** and a lower air discharge part **533** respectively, and then be discharge forward through the front surface panel **510**.

In the embodiment, the first fan module **540** and the second fan module **545** may have the same size and shape, for example. However, the first fan module **540** and the second fan module **545** may have a different size and shape.

The filter module **550** may be disposed between the suction surface and the fan module **540**, **545**. The filter module **550** includes a filter **559**, and the filter **559** filters air suctioned through the suction surface.

The filter **559** may form a filter surface that forms a flat surface in a direction orthogonal to the directions in which the fan suction and discharges air. That is, the filter **559** may form a filter surface parallel with the suction surface and the discharge surface. The suction surface, the filter surface of the filter **559** and the discharge surface may be disposed in a straight line.

Additionally, the filter **559** may form a filter surface parallel with the fan module **540**, **545**. The suction surface, the filter surface, the fan module **540**, **545** and the discharge surface may be disposed in a straight line.

The positions of the first fan module **540** and the second fan module **545** in the first direction and the second direction may correspond to the area occupied by the filter surface. Further, the surface areas taken up by the first fan module **540** and the second fan module **545** may correspond to the surface area of the filter surface. For example, when viewed from the front, the first fan module **540** and the second fan module **545** may be disposed to overlap the filter surface.

Furthermore, the directions of the first fan module **540** and the second fan module **545**'s suction of air may be the same as the directions of the suction surface and the discharge surface's disposition. That is, the first fan module **540** and the second fan module **545** may suction air respectively in the first direction or the axial direction.

The surface area occupied by the fan module **540**, **545** including the first fan module **540** and the second fan module **545** described above may correspond to the surface area of the suction surface and the surface area of the filter surface. Additionally, the surface area taken by the fan module **540**, **545** may correspond to the surface area of the discharge surface. Under the assumption that the suction surface, the filter surface and the discharge surface have a corresponding surface area, the surface area occupied by the fan module **540**, **545** may correspond to the surface area of each of the suction surface, filter surface and discharge surface.

Accordingly, air can be suctioned and discharged by the fan module **540**, **545** through the entire suction surface and the entire discharge surface.

For example, air can be suctioned by the fan module **540**, **545** through the suction surface that is ensured as much surface area as the fan module **540**, **545**. Air suctioned through the suction surface can be filtered by the filter surface that is ensured as much surface area as the suction surface and the fan module **540**, **545**. Additionally, air can be discharged by the fan module **540**, **545** through the discharge surface that is ensured as much surface area as the fan module **540**, **545**.

That is, the suction surface and the discharge surface may ensure a passage having an optimal surface area required for the fan module **540**, **545** to suction and discharge air. Additionally, the filter **559** may ensure a filter surface having

17

an optimal surface area required to filter air suctioned through the suction surface. Thus, air may flow effectively, based on the operation of the fan module 540, 545.

In the embodiment, the suction surface, the filter surface, the fan module 540, 545 and the discharge surface are disposed in parallel. Additionally, the suction surface, the filter surface, the fan module 540, 545 and the discharge surface are disposed in the first direction, and air flows in the same direction. That is, air, flowing based on the operation of the fan module 540, 545, can flow, in the same straight line, in the same direction as the direction in which the suction surface, the filter surface, the fan module 540, 545 and the discharge surface are disposed.

As air flows in a straight line, resistance against the flow of the air decreases, and the air can flow more smoothly. Accordingly, a sufficient amount of air is suctioned, and in response, a sufficient amount of air is discharged by the fan module 540, 545, leading to improvement in the air purification performance of the portable air purifier 50.

[Structure of Fan Module]

As described above, the portable air purifier 50 in the embodiment may include a plurality of fan modules 540, 545. In the embodiment, the portable air purifier 50 includes the first fan module 540 and the second fan module 545, and the first fan module 540 and the second fan module 545 have the same structure, for example. Herein, the structure of the first fan module 540 is described as an example.

FIG. 7 is a cross-sectional view specifically showing the structure of portion "VII" in FIG. 4, FIG. 8 is a perspective view separately showing a fan module of one embodiment, and FIG. 9 is an perspective view separately showing a fan case in FIG. 8. FIG. 10 is a front view separately showing a fan in FIG. 8, FIG. 11 is a side view showing the fan in FIG. 10, and FIG. 12 is a cross-sectional view along line "XII-XII" in FIG. 11.

In FIG. 7, some components of the portable air purifier are omitted.

Referring to FIGS. 3, 4, 7 and 8, the first fan module 540 may include a shaft 5410, a motor 5420, a fan case 5430 and a fan 5440.

The shaft 5410 may extend in the first direction, i.e., the axial direction. One side of the shaft 5410 in the axial direction may connect to the fan 5450. The other side of the shaft 5410 in the axial direction may connect to the fan case 5430. Detailed description in relation to this is provided hereafter.

The motor 5420 connects to the fan 5450 and applies a rotational force to the fan 5450. For example, the motor 5420 may be provided in the form of a BLDC motor in which a frequency can be adjusted. The motor 5420 may include a stator 5421 and a rotor 5423.

The stator 5421 may be disposed at the center side of the motor 5420 in the radial direction of the motor 5420, and the rotor 5423 may be disposed outside the stator 5421 in the radial direction of the stator 5421. That is, the motor 5420 in the embodiment may be provided in the form of an outer motor.

The stator 5421 may be fixed to the fan case 5430, 5440. The rotor 5423 may be fixed to the fan 5450. The rotor 5423 may rotate around the shaft 5410, outside the stator 5421, and the fan 5450 may rotate together with the rotor 5423 as the rotor 5423 rotates.

The fan case 5430, 5440 may support the stator 5421 and the shaft. The fan case 5430, 5440 may include a first support part 5430 and a second support part 5440, as illustrated in FIGS. 7 to 9.

18

The first support part 5430 may be disposed at the center side of the fan case 5430, 5440 in the radial direction of the fan case 5430, 5440. The first support part 5430 may include a support plate 5431 and a boss 5433.

The support plate 5431 may be formed into a circular plate. The support plate 5431 may form a flat surface that is parallel with at least any one of the suction surface, the filter surface and the discharge surface. The boss 5433 may protrude in the axial direction toward the fan 5450 from the support plate 5431. In the embodiment, the boss 5433 is disposed at the center of the support plate 5431 in the radial direction of the support plate 5431, for example.

The stator 5421 may be installed in the first support part 5430. The stator 5421 may be coupled to the boss 5433 while surrounding the boss 5433 from the outside in the radial direction. That is, the stator 5421 and the boss 5433 may be coupled in a way that the boss 5433 is fitted into the stator 5421 in the axial direction. The support plate 5431 may support the boss 5433 coupled to the stator 5421 at one side of the boss 5433 in the lateral direction, as described above.

Additionally, one side of the stator 5421 coupled to the boss 5433 in the axial direction of the stator 5421 may face the support plate 5431. While one side of the stator 5421 in the axial direction of the stator 5421 contacts the support plate 5431, the stator 5421 and the support plate 5431 may be coupled. Certainly, the stator 5421 may be installed in the first support part 5430 in a way that the stator 5421 is spaced from the support plate 5431.

Further, the boss 5433 may have a hollow hole, therein. The shaft 5410 may be inserted into the boss 5433 through the hollow hole. The shaft 5410 may pass through the boss 5433 in the axial direction and protrude from the boss 5433, and connect to a below-described hub 5451 of the fan 5450, outside the boss 5433.

A bearing 5435 may be inserted into the boss 5433 having the hollow hole. The shaft 5410 may be coupled to the bearing 5435, in the boss 5433. Accordingly, the shaft 5410 may be rotatably supported by the bearing 5435. In the embodiment, a pair of bearings 5435 is spaced a predetermined distance apart from each other in the axial direction, for example. The bearing 5435, disposed as described above, may help to install the shaft 5410 rotatably in the boss 5433 in a more reliable manner.

The second support part 5440 may support the first support part 5430, and couple the first fan module 540 to at least any one of the fan cover 530 and the filter module 550.

The second support part 5440 may be disposed outside the first support part 5430 in the radial direction of the first support part 5430. The second support part 5440 may support the first support part while surrounding the first support part 5430 from the outside in the radial direction.

The second support part 5440 may be formed into an approximate rectangle. For example, the length of the second support part 5440 in the second direction may be determined to correspond to the length of the filter module 550 in the second direction. Additionally, the length of the second support part 5440 in the third direction may be determined to correspond to half the length of the filter module 550 in the third direction. That is, if a pair of second support parts 5440 is disposed in the third direction, the exterior shapes of the pair of second support parts 5440 and the filter module 550 may be approximately aligned, when viewed from the front.

The second support part 5440 may have a hollow hole, therein. The hollow hole of the second support part 5440 may be formed in a way that the hollow hole penetrates the

inside of the second support part **5440** in the axial direction. The hollow hole may have a radius greater than a radius of the support plate **5431**. That is, a hollow hole bigger than the support plate **5431** is formed in the second support part **5440**, and the support plate **5431** may be disposed in the second support part **5440**.

Accordingly, the second support part **5440** may be spaced a predetermined distance apart from the first support part **5430**, specifically, from the support plate **5431** in the centrifugal direction.

A fan outlet **5430a** may be formed between the support plate **5431** and the second support part **5440** that are spaced as described above. The fan outlet **5430a** may form a passage allowing air having flown into the first fan module **540** to pass through the first fan module **540** and escape from the first fan module **540**.

The fan case **5430**, **5440** may further include a connection part **5445**. The connection part **5445** may extend from the support plate **5431** in the centrifugal direction and connect to the second support part **5440**. The connection part **5445** may connect between the support plate **5431** and the second support part **5440** and help the second support part **5440** to support the support plate **5431**.

In the embodiment, a plurality of connection parts **5445** may be disposed between the support plate **5431** and the second support part **5440**. Additionally, each of the connection parts **5445** may have a width less than the length of the connection part **5445** in the radial direction. Each of the connection parts **5445**, formed as described above, may be spaced at a predetermined interval along the circumferential direction of the support plate **5431**.

Accordingly, while the fan outlet **5430a** is formed between the support plate **5431** and the second support part **5440**, the fan outlet **5430a** may be respectively formed among the plurality of connection parts **5445**.

Additionally, the connection part **5445** may provide a passage through which an electric wire, connected to the stator **5421** on the support plate **5431**, passes. The electric wire connecting to the stator **5421** may be withdrawn toward the second support part **5440** side through the area where the connection part **5445** is disposed. At this time, the electric wire may pass through the fan outlet **5430a** area in the state being covered by the connection part **5445**, and connect between the stator **5421** and the main PCB **595** (see FIG. 3). When viewed from the front, the electric wire that is withdrawn from the stator **5421** and passes through the fan outlet **5430a** area is cover by the connection part **5445** and is not be seen.

The second support part **5440** may be provided with a side rib **5443**. The side rib **5443** may be disposed at at least any one of both sides of the second support part **5440** in the second direction of the second support part **5440**. The side rib **5443** may protrude from a lateral portion of the second support part **5440** in the second direction of the second support part **5440**.

The length of the side rib **5443** in the second direction may be set to correspond to the length of the second support part **5440** in the second direction. Additionally, the length of the side rib **5443** in the first direction may be less than the length of the first support part **5430** in the first direction. That is, the side rib **5443** may have a thickness less than that of the first support part **5430**. For example, the side rib **5443** may have a thickness that is one fourth or less of the thickness of the first support part **5430**.

The side rib **5443** may be biased toward the front or the rear of the second support part **5440** while protruding from

the second support part **5440**. In the embodiment, the side rib **5443** is biased toward the rear of the second support part **5440**, for example.

As the fan module **540**, **545** is inserted into the case **520**, the side rib **5443** may contact both lateral surfaces of the case **520**, i.e., the inner surfaces of the first surface parts **521**. Accordingly, predetermined space may be formed between a lateral surface of the second support part **5440** and the first surface part **521**.

The formed space may be used as space for allowing an electric wire to pass. For example, an electric wire connecting to the sub PCB **595** disposed on the fan module **540**, **545** may pass through the space and connect to the stator **5421** of the motor **5420**.

Referring to FIGS. 7 and 10 to 12, the fan **5450** may include a hub **5451** and fan blade **5455**.

The hub **5451** is disposed at the center of the fan **5450** in the radial direction of the fan **5450**, and rotates together with the rotor **5423** and the shaft. The hub **5451** may include a first fan module **540**, **545** and a skirt part **5433**.

The first fan module **540**, **545** may be formed into a circular plate parallel with the support plate **5431**. The first fan modules **540**, **545** may be disposed in parallel with the support plate **5431** with the boss **5433** between the first fan modules **540**, **545**.

The first fan module **540**, **545** may be provided with a shaft coupling part **5452a**. The shaft coupling part **5452a** may be disposed at the center of the first fan module **540**, **545** in the radial direction of the first fan module **540**, **545**. The shaft coupling part **5452a** may protrude in the axial direction toward the boss **5433** from the first fan module **540**, **545**.

The shaft coupling part **5452a** may be coupled to the end portion of the shaft **5410** in the axial direction of the shaft **5410**. For example, the shaft coupling part **5452a** may be coupled to the shaft **5410** in a way that the shaft **5410** is fitted into the shaft coupling part **5452a**. The shaft **5410** may be fixed to the shaft coupling part **5452a** or rotatably coupled to the shaft coupling part **5452a**.

The skirt part **5453** may protrude toward the support plate **5431** from the edge of the hub plate part **5452**. The skirt part **5453** may form a slant surface that inclines in the centrifugal direction as the skirt part **5452** becomes farther from the hub plate part **5452** in the axial direction. For example, the shape in which the hub plate part **5452** and the skirt part **5453** connect may be a truncated cone shape which has a hollow hole therein and one side of which is open. The skirt part **5433** may be disposed outside the stator **5421** in the radial direction of the stator **5421**. That is, the stator **5421** may be disposed in space surrounded by the skirt part **5453** and the hub plate part **5452**.

The fan blade **5455** may protrude from the hub **5451** in the centrifugal direction. The fan **5450** may be provided with a plurality of fan blades **5455**, and the fan blades **5455** may be spaced a predetermined distance apart from one another along the circumference direction of the hub **5451**.

Specifically, the fan blade **5455** may protrude from the skirt part **5453** in the centrifugal direction. At this time, the inside of the fan blade **5455** in the radial direction thereof may connect to the skirt part **5453**, and the outside of the fan blade **5455** in the radial direction thereof may connect to a shroud **5457** that is described hereafter. That is, the skirt part **5453** is a portion of the hub **5451**, which directly connects to the fan blade **5455** and directly contacts air passing through the first fan module **540**. The skirt part **5453** may closely relate to a flow path of air passing through the first fan module **540**.

21

The fan 5450 may further include a shroud 5457. The shroud 5457 may be spaced a predetermined distance apart from the hub 5451 in the radial direction, outside the hub 5451 in the radial direction thereof. The shroud 5457 may be spaced from the hub 5451 by a distance corresponding to the length of the fan blade 5455 in the radial direction of the fan blade 5455. Additionally, each of the fan blades 5455 may connect between the hub 5451, specifically, the skirt part 5453 and the shroud 5457.

The shroud 5457 may form a slant surface that inclines in the centripetal direction as the shroud 5457 becomes farther from the fan case 5430, 5440 in the axial direction, i.e., toward the rear. For example, the shroud 5457 may form a slant surface that is approximately parallel with the skirt part 5453. In the embodiment, a gap between the skirt part 5453 and the shroud 5457 increases further toward the front, for example.

Each of the fan blades 5455, connecting between the shroud 5457 and the skirt part 5433, may include a leading edge 5455a, a trailing edge 5455b, a shroud chord 5455c and a hub chord 5455d.

The leading edge 5455a may be disposed at the front end of the fan 5450 in a rotation direction and formed in a straight manner. The rotation direction is defined as a direction in which the fan 5450 rotates. The leading edge 5455a may be formed into a straight line that is disposed at the front end of the fan 5450 in the rotation direction of the fan 5450 and extends in the radial direction.

The trailing edge 5455b may be disposed at the rear end of the fan in the rotation direction and formed in a straight manner. The trailing edge 5455b may be formed into a straight line that extends in a direction between the axial direction and the radial direction.

The shroud chord 5455c may connect one end of the leading edge 5455a and one end of the trailing edge 5455b. The shroud chord 5455c may extend from the inner circumferential surface of the shroud 5457.

The hub chord 5455d may connect the other end of the leading edge 5455a and the other end of the trailing edge 5455b. The hub chord 5455d may extend from the outer circumferential surface of the hub 5451.

Additionally, one end of the leading edge 5455a and one end of the trailing edge 5455b may connect to the inner circumferential surface of the shroud 5457. The other end of the leading edge 5455a and the other end of the trailing edge 5455b may connect to the outer circumferential surface of the skirt part 5453.

One end of the leading edge 5455a may be disposed closer to the center of the hub plate part 5452 in the radial direction of the hub plate part 5452 than one end of the trailing edge 5455b. The other end of the leading edge 5455a may be disposed closer to the center of the hub plate part 5452 in the radial direction of the hub plate part 5452 than the other end of the trailing edge 5455b, since one end and the other end of the leading edge 5455a is disposed further forward than one end and the other end of the trailing edge 5455b in the rotation direction, and the radius of the skirt part 5453 decreases further toward the front in the rotation direction.

Additionally, the hub 5451 may include an inner projection part 5454. The inner projection part 5454 may protrude toward the support plate 5431 from the hub plate part 5452. In the embodiment, the inner projection part 5454 and the skirt part 5453 protrude from the same point on the hub plate part 5452, e.g., from the edge of the hub plate part 5452, for example.

22

The inner projection part 5454 may be formed in a way that the inner projection part 5454 extends from the hub plate part 5452 in the axial direction. For example, the shape in which the hub plate part 5452 and the skirt part 5453 connect may be a cylinder shape which has a hollow hole therein and one side of which is open.

The inner projection part 5454 may be disposed between the skirt part 5453 and the stator 5421. That is, the stator 5421 may be disposed in space surrounded by the inner projection part 5454 and the hub plate part 5452.

The rotor 5423 may be disposed between the inner projection part 5454 and the stator 5421. The rotor 5423 may be fixed onto the inner circumferential surface of the inner projection part 5454. That is, the stator 5421 may be fixed to the fan case 5430, 5440, and the rotor 5423 may be fixed to the fan 5450. Accordingly, the fan 5450 may rotate together with the rotor 5423 as the rotor 5423 rotates.

In the embodiment, the fan blade 5455 may connect to the skirt part 5453 of the hub 5451. To induce the flow of air flowing into the fan module 540, 545 in a direction between the axial direction and the radial direction, the skirt part 5453 forms a slant surface that inclines in a direction between the axial direction and the radial direction.

Since the skirt part 5453 is formed at a slant, it is difficult to fix the rotor 5423 to the inner circumferential surface of the skirt part 5453. For example, for the rotor 5423 to be fixed to the inner circumferential surface of the skirt part 5453, the rotor 5423 needs to be entirely formed into an approximate truncated cone having a hollow hole since the shape of the outer circumferential surface of the rotor 5423 needs to correspond to the shape of the inner circumferential surface of the skirt part 5453. However, the shape of the rotor 5423 may be inappropriate for the reliable driving of the motor 5420.

Considering the fact, in the embodiment, the inner projection part 5454 is provided between the skirt part 5453 and the stator 5421. The inner projection part 5454 may provide a fixation surface that allows the rotor 5423 to be fixed stably to the inside of the hub 5451 when the rotor 5423 is entirely formed into a cylinder having a hollow hole.

Additionally, the inner projection part 5454 may form a structure for improving the rigidity of the hub 5451, in the hub 5451. The inner projection part 5454 may help to improve the rigidity of the entire fan 5450 effectively while suppressing a significant increase in the weight of the entire fan 5450.

[Structure of Coupling Between Fan Module and Fan Cover]

FIG. 13 is an exploded perspective view separately showing a fan cover and a fan module.

Referring to FIG. 13, the fan cover 530 may include an upper cover part 531 and a lower cover part 535.

The upper cover part 531 is disposed at the front of the fan module 540, 545. The upper cover part 531 may be provided with an air discharge part 532, 533. The air discharge part 532, 533 may be formed in a way that a portion of the upper cover part 531 penetrates or is cut. The air discharge part 532, 533 may form a passage that connects between the front of the case 520, i.e., the discharge surface, and the fan 5450 of the fan module 540, 545, on the fan cover 530.

In the embodiment, the portable air purifier 50 is provided with two fan modules 540, 545, for example. That is, the first fan module 540 and the second fan module 545 are disposed vertically in the accommodation space of the case 520.

Accordingly, the fan cover 530 may also be provided with two air discharge parts 532, 533. That is, an upper air discharge part 532 and a lower air discharge part 533 may be disposed vertically at the upper cover part 531.

Air having passed through the first fan module **540** is discharged forward through the upper air discharge part **532**, and air having passed through the second fan module **545** may be discharged forward through the lower air discharge part **533**.

In another example, the portable air purifier may be provided with one fan assembly or three or more fan assemblies, and accordingly, the fan cover may be provided with one air discharge part or three or more air discharge parts.

Additionally, the upper cover part **531** may be provided with a first fastening projection **534**. The first fastening projection **534** may protrude from the rear surface of the upper cover part **531** rearward. In response, the fan case **5430**, **5440** may be provided with a fastening hole **5441**. The fastening hole **5441** may be formed in a way that penetrates in the front-rear direction in the second support part **5440**.

In an example, a total of four fastening holes **5441** may be provided, and each of the fastening holes **5441** may be disposed at each edge of the second support part **5440**. The first fastening projection **534** may be disposed respectively in positions corresponding to the positions of the fastening holes **5441**.

Each of the first fastening projections **534** may be inserted into the fastening hole **5441** and fitted into and coupled to the second support part **5440**. As a result of the coupling between the first fastening projection **534** and the second support part **5440**, the fan cover **530** and the fan module **540**, **545** may be coupled at a plurality of points.

As the fan cover **530** and the fan module **540**, **545** are coupled, the fan module **540**, **545** may be fixed to the rear of the fan cover **530**. At this time, the fan module **540**, **545** may be fixed in the position where the fan outlet **5430a** formed at the front of the fan module **540**, **545** overlaps the air discharge part **532**, **533** formed at the fan cover **530** in the first direction. Thus, a straight line passage may be formed to allow air discharged from the fan module **540**, **545** to pass through the fan cover **530**.

[Structure of Fan Base]

FIG. **14** is an exploded perspective view separately showing a fan module, a fan base and a filter, FIG. **15** is a rear perspective view showing the rear surface of a fan base, and FIG. **16** is a perspective view showing a coupling state between the fan base and the filter.

Referring to FIGS. **7** and **13** to **16**, the portable air purifier **50** according to the present disclosure may further include a fan base **5460**.

The fan base **5460** may be disposed between the filter module **550** and the fan module **540**, **545**. The fan base **5460** may be formed into a shape corresponding to the shape of the filter surface. For example, the fan base **5460** may be formed into a shape of the filter **559** viewed from the front, i.e., a rectangular shape. The fan base **5460** may include a base plate **5461**, and a bell mouth **5463**.

In this embodiment, two fan bases **5460** disposed in the third direction, i.e., the up-down direction, are disposed between the filter module **550** and the fan module **540**, **545**, for example. At this time, one fan base **5460** disposed in the upper portion is disposed between the first fan module **540** and the filter module **550**, and the other fan base **5460** disposed in the lower portion is disposed between the second fan module **545** and the filter module **550**.

When the fan base **5460** is provided separately depending on the number of the fan modules, the fan base **5460** may be provided in response to the number of the fan modules even if the number of the fan modules varies. That is, when one fan module is provided, one fan base **5460** is applied, and

when two or more fan modules are provided, the same number of the fan bases **5460** as the number of the fan modules may be stacked in the up-down direction. Thus, the fan base **5460** is provided regardless of the number of the fan modules, ensuring ease of management of components.

Despite the fact, the fan base **5460** may be provided in a way that a single fan base **5460** includes a plurality of fan inlets **5462**.

The base plate **5461** may be disposed between the filter module **550** and the fan module **540**, **545**. The length of the base plate **5461** in the first direction may be much less than that of the filter module **550** and the fan module **540**, **545**. For example, the base plate **5461** may be formed into a rectangular plate.

The base plate **5461** may have a fan inlet **5462**. The fan inlet **5462** may be formed in a way that fan inlet **5462** penetrates on the base plate **5461** in the first direction. The fan inlet **5462** may be disposed approximately in the position where the fan inlet **5462** overlaps the air discharge part **532**, **533** (see FIG. **13**) of the fan cover **530** and the fan outlet **5430a** of the fan module **540**, **545** in the first direction. The fan inlet **5462** formed as described above may form a passage that connects between the filter **559** and the fan module **540**, **545**, at the fan base **5460**.

The bell mouth **5463** may protrude from the base plate **5461**. The bell mouth **5463** may protrude from the base plate **5461** to the fan module **540**, **545**, in the axial direction.

In the embodiment, the fan inlet **5462** has the same shape as the shroud **5457**, i.e., a circular shape, for example. Additionally, the radius of the fan inlet **5462** may be set similarly to the radius of a portion of the shroud **5457**, which is adjacent to the fan inlet **5462**.

The bell mouth **5463** may be formed in a way that the bell mouth **5463** surrounds the outer circumferential surface of the fan inlet **5462** formed as described above. In other words, the fan inlet **5462** may be formed in a way that the fan inlet **5462** penetrates at the inside of the bell mouth **5463** in the radial direction of the bell mouth **5463**.

The bell mouth **5463** may protrude toward the fan module **540**, **545** from the fan inlet **5462** in the first direction. At this time, at least a portion of the bell mouth **5463** may be inserted into the shroud **5457** in the radial direction of the shroud **5457**. The bell mouth **5463** guides a suction flow at the entrance of the fan module **540**, **545**, and help to improve the suction and discharge performance of the fan module **540**, **545**.

The fan base **5460** may be coupled to the fan base **5460**. To this end, any one of the fan base **5460** and the filter module **550** may have a fastening boss **5433**, and any one of the fan base **5460** and the filter module **550** may have a second fastening projection **553**. In the embodiment, the fan base **5460** has the fastening boss **5433**, and the filter module **550** has the second fastening projection **553**, for example.

The fastening boss **5433** may protrude from the base plate **5461** to the filter module **550** in the first direction. The fastening boss **5433** may have a hollow hole. The second fastening projection **553** may protrude from the filter module **550**, specifically, the front surface of the filter case **551**, toward the fan base **5460** in the first direction. The second fastening projection **553** may be fitted into and coupled to the fastening boss **5433** in a way that the second fastening projection **553** is inserted into the hollow hole of the fastening boss **5433**.

In an example, a total of four fastening bosses **5433** are provided, and each of the fastening bosses **5433** may be disposed at each edge of the first support part **5430**. The

second fastening projection **553** may be disposed respectively in positions corresponding to the positions of the fastening bosses **5433**.

Based on the coupling between the second fastening projection **553** and the fastening boss **5433**, the fan base **5460** and the filter module **550** may be coupled at a plurality of points. As the fan base **5460** is coupled to the filter module **550** as described above, the fan base **5460** may be fixed to the front of the filter module **550**.

Additionally, the fan base **5460** may be coupled to the fan module **540**, **545**. To this end, any one of the fan base **5460** and the fan module **540**, **545** is provided with a third fastening projection **5467**, and the other may be provided with a projection boss **5442**. In the embodiment, the fan base **5460** is provided with the third fastening projection **5467**, and the fan module **540**, **545** is provided with the projection boss **5442**, for example.

The projection boss **5442** may protrude from the second support part **5440** toward the fan base **5460** in the first direction. The projection boss **5442** may have a hollow hole. The third fastening projection **5467** may protrude from the front surface of the base plate **5461** toward the front. The third fastening projection **5467** may be fitted into and coupled to the fastening boss **5433** in a way that the third fastening boss **5467** is inserted into the hollow hole of the projection boss **5442**.

As the projection boss **5442** and the third fastening projection **5467** are coupled, the fan base **5460** and the fan module **540**, **545** may be coupled such that the second support part **5440** is spaced a predetermined distance apart from the base plate **5461**. At this time, the predetermined distance is set to the length of a portion of the fan **5450** in the axial direction, protruding outward in the radial direction of the second support part **5440**, e.g., the length of the shroud **5457** in the first direction, or greater.

Further, a total of four projection bosses **5442** are provided, and each of the projection bosses **5442** may be disposed at each edge of the second support part **5440**. The third fastening projection **5467** may be disposed respectively in positions corresponding to the positions of the projection bosses **5442**.

A gap between the portions where each of the projection bosses **5442** and each of the third fastening projections **5467** are coupled may be open in the radial direction. Additionally, a portion of the fan **5450**, specifically, a portion of the shroud **5457**, may protrude outward in the radial direction of the second support part **5440** through the open portions.

The fan base **5460** described above may serve as a coupling medium for coupling between the fan module **540**, **545** and the filter module **550**, and guide a suction flow at the entrance of the fan module **540**, **545** to improve the suction and discharge performance of the fan module **540**, **545**.
[Aspect of Air Flow of Portable Air Purifier]

FIG. 17 is a view showing an aspect of the air flow of the portable air purifier of one embodiment.

In FIG. 17, some components are omitted from the portable air purifier.

Hereafter, an aspect of the air flow of the portable air purifier in the embodiment is described with reference to FIGS. 4 and 17.

Referring to FIGS. 4 and 17, as the fan module **540**, **545** operates, air behind the portable air purifier **50** flows into the portable air purifier **50**. At this time, the air behind the portable air purifier **50** may pass through the suction surface through the first suction inlet **570a** formed on the rear surface cover **580**.

The air having passed through the suction surface and having flown into the portable air purifier **50** passes through the filter **559**, and while the air passes through the filter **559**, the filter **60** can filter physical particles such as dust/fine dust/ultra fine dust and the like, chemical substances such as odorant particles/harmful gases and the like, and microorganisms such as germs/viruses and the like, that are included in the air.

At this time, the air may be suctioned into the filter **559** through the suction surface that has as much surface area as the filter surface. Additionally, the air suctioned through the suction surface may be filtered through the filter surface that has as much surface area as the fan module **540**, **545**.

That is, air may be suctioned and filtered effectively through the suction surface and the filter surface that have sufficient surface areas corresponding to the surface area of the fan module **540**, **545**. Additionally, since the suction surface, the filter surface and the fan module **540**, **545** are disposed in a straight line, air may be suctioned and filtered effectively while flow loss is minimized.

The air having passed through the filter **559**, i.e., purified air, may flow into the fan module **540**, **545** through the fan inlet **5462**. The flow of the air passing through the fan inlet **5462** may be guided by the bell mouth **5463**, and accordingly, a smooth inflow of air to the fan module **540**, **545** may be induced effectively.

The air having flown into the fan module **540**, **545** may be discharged in a mixed flow direction while being discharged from the front of the fan module **540**, **545** through the fan outlet **5430a**. The mixed flow direction may be defined as a front-side diagonal direction.

The air, discharged from the front of the fan module **540**, **545**, i.e., purified air, passes through the fan cover **530**, and is discharged from the front of the portable air purifier **50**. At this time, the purified air may pass through the discharge surface through the discharge outlet **510a** formed on the front cover **510**.

The purified air may be discharged through the discharge surface that has as much surface area as the suction surface, the filter surface and the fan module **540**, **545**. That is, the purified air may be discharged effectively through the discharge surface that has a sufficient surface area corresponding to the surface area of the suction surface, the surface area of the filter surface and the surface area of the fan module **540**, **545**. Further, since the suction surface, the filter surface, the fan module **540**, **545** and the discharge surface are disposed in a straight line, air may be suctioned and filtered, and purified air may be discharged, effectively, while flow loss is minimized.

[Effect of Portable Air Purifier with Mixed Flow Fan-Type Fan]

The portable air purifier **50** in the embodiment include a mixed flow fan type-fan **5450**. The portable air purifier **50** may produce the following effects.

First, since the fan module **540**, **545** includes a mixed flow fan-type fan **5450**, the fan module **540**, **545** may ensure improvement in air purification performance, compared to a fan module including an axial flow fan-type fan.

The axial flow fan allows air to be suctioned and discharged in a straight line. The axial flow fan may have a small thickness, and can be applied to a small-sized air purifier.

However, the performance of the axial flow fan significantly deteriorates at fixed pressure. For example, if the density or the thickness of the filter **559** increases to enhance filtering performance, the suction and discharge performance of the axial flow fan may significantly deteriorate.

Accordingly, it is difficult to apply an axial flow fan to a portable air purifier with a high performance filter.

The mixed flow fan is somewhere between an axial flow fan and a centrifugal fan, and ensures more excellent performance at fixed pressure than the axial flow fan on condition that the mixed flow fan has the same size as the axial flow fan. However, the length of a mixed flow fan in the axial direction is greater than that of an axial flow fan on condition that the mixed flow fan has the same size as the axial flow fan.

However, the axial flow fan is inappropriate to send high-pressure high-speed air. Accordingly, a portable air purifier to which an axial flow fan is applied needs to further include a structure such as a guide vane. For example, a guide vane may be formed in the air discharge part **532**, **533** of the fan cover **530**. The guide vane may be a cause for an increase in the length of the fan cover **530** in the first direction, i.e., the thickness of the fan cover **530**.

An increase in the thickness of the fan cover **530**, caused by the guide vane, results in a decrease in the length of the axial flow fan in the first direction due to the limited space in the portable air purifier, and a deterioration in the suction and discharge performance of the axial flow fan.

Unlike an axial flow fan, a mixed flow fan is appropriate to send high-pressure high-speed air. That is, without a structure such as a guide vane, the mixed flow fan may send air much farther than the axial flow fan.

For this reason, in the embodiment, the fan cover **530** excludes a structure such as a guide vane. Accordingly, the thickness of the fan cover **530** decreases, and the fan module **540**, **545** is small enough to be inserted into the air discharge part **532**, **533**. For example, the length of the fan module **540**, **545** in the first direction may be determined to the extent that the front surface of the fan module **540**, **545** and the front surface of the fan cover **530** constitute the same surface. That is, the length of the fan module **540**, **545** in the first direction may extend by a length occupied by a guide vane.

As the size of the fan module **540**, **545** increases, the suction and discharge performance of the fan module **540**, **545** improves, enabling the portable air purifier **50** in the embodiment to ensure improvement in air purification performance.

Additionally, in the structure of the fan module **540**, **545**, the fan case **5430**, **5440** does not surround the fan **5450** from the outside in the radial direction. That is, the fan **5450** may be formed to protrude further in the centrifugal direction than the inner circumferential surface of the second support part **5440**. Accordingly, at least a portion of the fan **5450** may protrude to a gap between the second support part **5440** and the base plate **5461**.

Since the fan **5450** protrudes further in the centrifugal direction than the inner circumferential surface of the second support part **5440** as described above, the size of the fan **5450** may increase.

As the size of the fan module **540**, **545** increases as described above, the suction and discharge performance of the fan module **545**, **545** improves, enabling the portable air purifier **50** in the embodiment to ensure improvement in air purification performance.

Second, since the fan module **540**, **545** includes a mixed flow fan-type fan **5450**, the range in which the portable air purifier **50** discharges air may expand.

A mixed flow fan can discharge air in the mixed flow direction, and accordingly, purified air discharged through

the discharge surface may be discharged in the mixed flow direction, i.e., a direction between the front and the centrifugal direction.

When it comes to a small-sized portable air purifier **50**, if purified air is discharged toward the front only in a straight line, the range of the discharge of the purified air is limited to a range corresponding to the discharge surface.

If the purified air is discharged within a range corresponding to the discharge surface, even considering the properties of the small-sized portable air purifier **50**, limitations are imposed on the use of the portable air purifier **50**.

For example, the user has to keep holding up the portable air purifier **50** or has to accurately adjust the position of the portable air purifier **50** such that the discharge outlet **510a** faces the user's face, causing inconvenience to the user.

The portable air purifier **50** in the embodiment may discharge high-pressure high-speed air in a direction between the front and the centrifugal direction. Accordingly, purified air may spread to a large area as well as reaching a far corner, enabling the portable air purifier **50** to ensure improvement in air purification performance.

Third, each fan module **540**, **545** includes the mixed flow fan-type fan **5450**, and a plurality of fan modules **540**, **545** are stacked, allowing purified air to be discharged far away.

As described above, a mixed flow fan can discharge air in the mixed flow direction. At this time, a collision between air discharged from the first fan module **540** and air discharged from the second fan module **545** may occur, in the portion where the fan modules **540**, **545** are adjacent to each other.

For example, air may be discharged downward at a slant at the lower side of the first fan module **540** adjacent to the second fan module **545**, and air may be discharged upward at a slant at the upper side of the second fan module **545** adjacent to the first fan module **540**.

At this time, a portion of the air discharged from the first fan module **540** and a portion of the air discharged from the second fan module **545** may collide with each other. Thus, their radius wise speed vector decreases, and their front wise speed vector may increase.

A decrease in the radius wise speed vector and an increase in the front wise speed vector denote the direction of speed vector of air discharged from the portable air purifier **50** changes to a direction close to the front.

As a result, the portable air purifier **50** discharges air mainly toward the front, and allows the air to reach a far corner, ensuring improvement in air purification performance.

The embodiments are described above with reference to a number of illustrative embodiments thereof. However, the embodiments set forth herein are provided only as examples, and numerous other modifications and embodiments can be devised by one skilled in the art without departing from the technical spirit of the disclosure. The technical protection scope of the disclosure should be defined according to the appended claims.

DESCRIPTION OF REFERENCE NUMERALS

- 50:** Portable air purifier
- 510:** Front surface panel
- 520:** Case
- 530:** Fan cover
- 531:** Upper cover part
- 532:** Upper air discharge part
- 533:** Lower air discharge part
- 534:** First fastening projection
- 535:** Lower cover part

29

540,545: Fan module
540: First fan module
545: Second fan module
550: Filter module
551: Filter case
552: Penetration hole
553: Second fastening projection
559: Filter
560: Battery
570: Rear surface panel
580: Rear surface cover
5410: Shaft
5420: Motor
5421: Stator
5423: Rotor
5430, 5440: Fan case
5430: First support part
5430a: Outlet
5431: Support plate
5433: Boss
5435: Bearing
5440: Second support part
5441: Fastening hole
5442: Projection boss
5443: Side rib
5445: Connection part
5450: Fan
5451: Hub
5452: Hub plate part
5452a: Shaft coupling part
5453: Skirt part
5454: Inner projection part
5455: Fan blade
5455a: Leading edge
5455b: Trailing edge
5455c: Shroud chord
5455d: Hub chord
5457: Shroud
5460: Fan base
5461: Base plate
5462: Fan inlet
5463: Bell mouth
5465: Fastening boss
5467: Third fastening projection

The invention claimed is:

1. A portable air purifier, comprising:

- a case having a suction surface on a first side thereof in an axial direction and having a discharge surface on a second side thereof in the axial direction;
- a filter disposed between the suction surface and the discharge surface;
- a fan module that is disposed between the discharge surface and the filter, suctions air, having passed through the filter, in the axial direction, and discharges the air in a direction between the axial direction and a radial direction, the fan module comprising a mixed flow fan; and

30

a fan base disposed between the filter and the fan module, wherein the fan base comprises:

a fan inlet that forms a passage that connects between the filter and the fan module;

a base plate disposed between the filter and the fan module; and

a bell mouth that protrudes from the base plate toward the fan module in the axial direction, wherein the fan module comprises:

a motor including a stator and a rotor that is disposed outside of the stator in a radial direction of the stator; and

a fan case that supports at least a portion of the motor, wherein the fan case comprises:

a first support that supports the stator and the shaft;

a second support that supports the first support while surrounding the first support from the outside in the radial direction;

a projection boss that is provided at the second support and configured to be coupled to the fan base; and

a fan outlet that forms a passage through which air is discharged from the fan module, wherein the fan base further comprises a fastening projection that protrudes from the base plate toward the fan module in the axial direction, wherein the projection boss protrudes from the second support toward the base plate in the axial direction, wherein at least a portion of the fan protrudes outward in a radial direction of the second support, wherein as the projection boss and the fastening projection are coupled, the fan base and the fan module are coupled such that the second support and the base plate are spaced a predetermined distance apart from each other, wherein the predetermined distance is a length or greater of the portion of the fan, protruding outward in the radial direction of the second support, in the axial direction, wherein the fan module comprises a first fan module disposed in parallel with the discharge surface, between the discharge surface and the filter, and a second fan module disposed on a same surface as the first fan module, wherein each of the first fan module and the second fan module is disposed in parallel with the discharge surface, and wherein a fan outlet of the first fan module and a fan outlet of the second fan module are disposed on a same surface.

2. The portable air purifier of claim 1, wherein the fan inlet is formed at an inside of the bell mouth in such a way that the fan inlet penetrates in a radial direction of the bell mouth.

3. The portable air purifier of claim 1, wherein the fan case is coupled to the fan base, and the fan base is coupled to the filter.

* * * * *